PathScribe by ForMedrix AI

Administrator Guide

System Administration, Configuration & Report Template Management

For Administrators and Lead Pathologists

Combined Edition -- Part I: Administration | Part II: Report Template Configuration

June 2026 | Confidential -- PathScribe AI

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Part I -- Administration

System administration and configuration. For administrators and lead pathologists.

1. Administration Overview

Administrators access PathScribe from the Home screen via the Configuration tile, divided into four tabs: System, Staff, Protocols, and (when HL7/FHIR is active) LIS Integration.

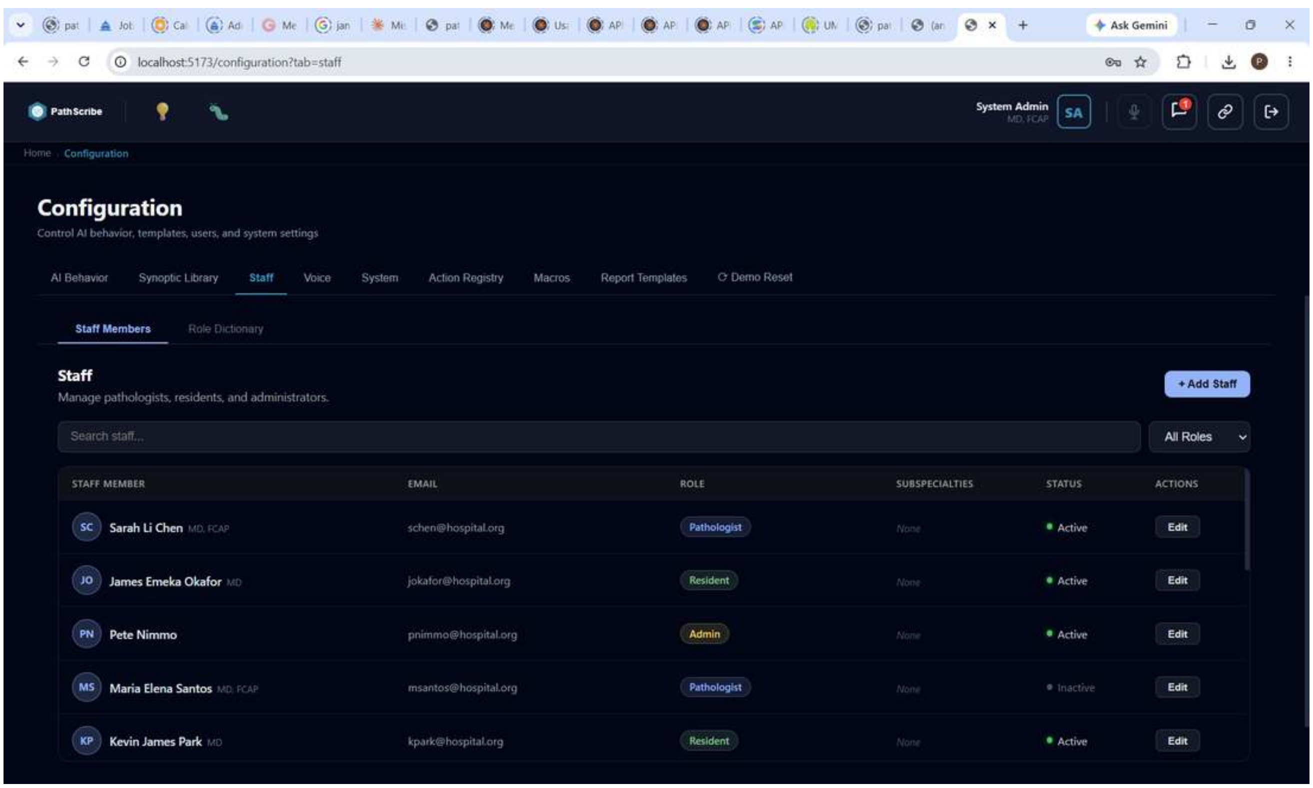
> **IMPORTANT**
>
> The Admin role provides configuration access only. Admins cannot see clinical case data by design. A user who needs both clinical and admin access must hold both the Pathologist and Admin roles simultaneously.

PathScribe operates in two deployment modes:

Mode	Description	Typical customer
Co-Pilot	AI assistance and synoptic reporting alongside your existing LIS. The LIS owns case creation and status.	Hospital labs running CoPath, Epic Beaker, Sunquest, etc.
Standalone LIS	PathScribe is the system of record -- case management, gross/micro entry, report assembly, sign-out, and full coding.	Independent pathology groups without existing AP-LIS.

2. Staff Management

PathScribe distinguishes two categories of people -- Staff (internal users who log in) and Physicians (external ordering doctors who never log in).

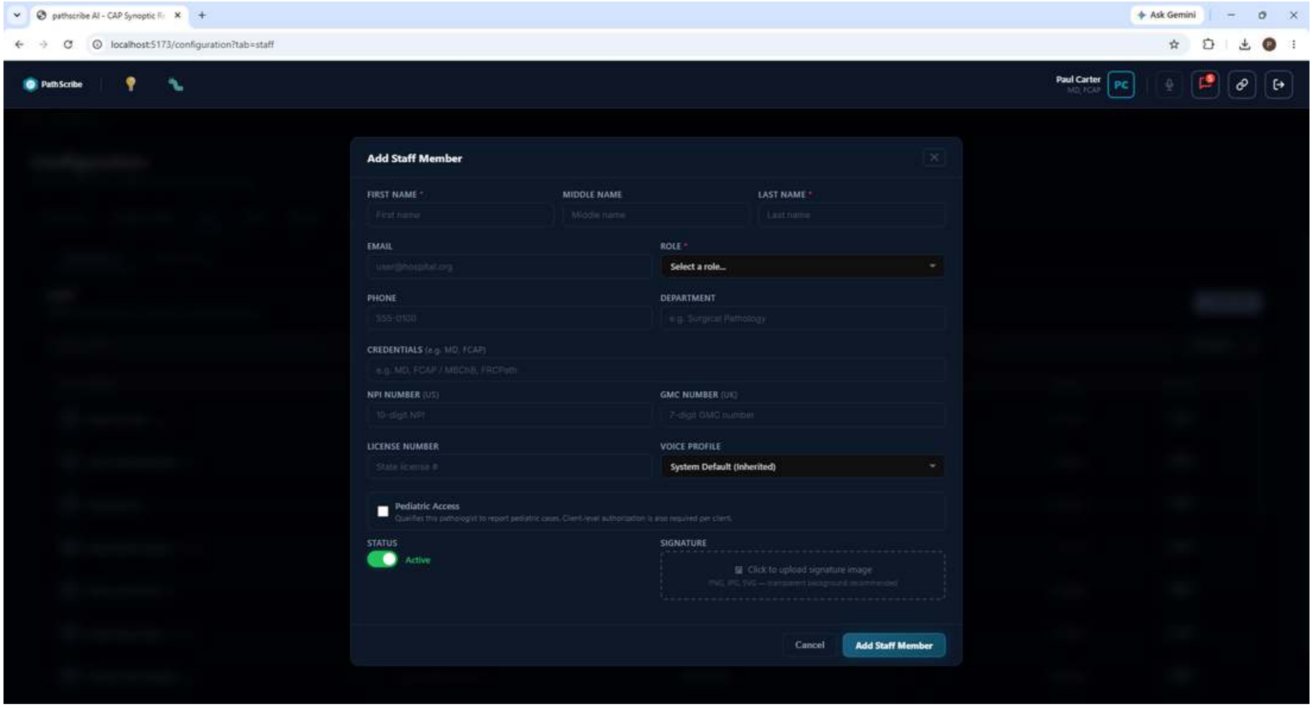


Staff Members -- name, email, role, subspecialties, and status.

Category	Who they are	Log in?	Managed in
Staff	Pathologists, Residents, Administrators	Yes -- login credentials	Configuration -> Staff
Physicians	External ordering / referring doctors	Never -- directory only	Configuration -> System -> Physicians

2.1 Adding a New Staff Member

- Navigate to Configuration -> Staff and click + Add Staff.
- Complete the required fields (see table below). Click Add Staff Member.
- The new user can log in immediately with their email address.



Add Staff Member -- full field set, including Credentials, NPI/GMC, and Signature upload.

Field	Required	Notes
First / Middle / Last Name	First & Last	Appears on reports, audit logs, and in-app display.
Email	No	Used for login (if SSO not configured) and notifications.
Role	Yes	Single-select here; permissions are the union of all roles a user ends up assigned.
Phone	No	Used for directory and urgent notification routing.
Department	No	e.g. Surgical Pathology.
Credentials	No	e.g. MD, FCAP / MBChB, FRCPath -- distinct from License Number below.
NPI Number (US)	No	10-digit. Required for billing-related workflows.
GMC Number (UK)	No	7-digit General Medical Council number.
License Number	No	State/jurisdiction medical licence -- separate from Credentials.

Field	Required	Notes
Voice Profile	No	Defaults to System Default (Inherited); can be overridden per user.
Pediatric Access	No	Qualifies this pathologist to report pediatric cases. Client-level authorization is also required per client -- this checkbox alone is not sufficient.
Status	Yes	Active / Inactive. Inactive users cannot log in.
Signature	No	Upload an image (PNG/JPG/SVG, transparent background recommended) for report sign-off.

2.2 Roles and Permissions

Roles are not mutually exclusive. Permissions are the union of all assigned roles.

Role	Case worklist	Sign-out	Configuration	Notes
Pathologist	Full access	Primary sign-out	None	Core clinical role. Full case lifecycle.
Resident	Full access	Co-sign only	None	Sign-out gating controlled by usePermission config.
Admin	None	None	Full access	Cannot access clinical case data by design.

2.3 Deactivating and Reactivating Users

Click a user row to expand their record, then click Deactivate. The account is preserved for audit purposes but login is blocked. Click Reactivate to restore access.

2.4 Paediatric Access Control

Cases involving patients below the configured paediatric age threshold are only visible to users with Paediatric Access enabled. Users without this flag see the case row with all PHI redacted.

3. Physician Directory

Navigate to Configuration -> System -> Physicians. Physicians appear on reports and drive result delivery routing. They do not log in and hold no permissions.

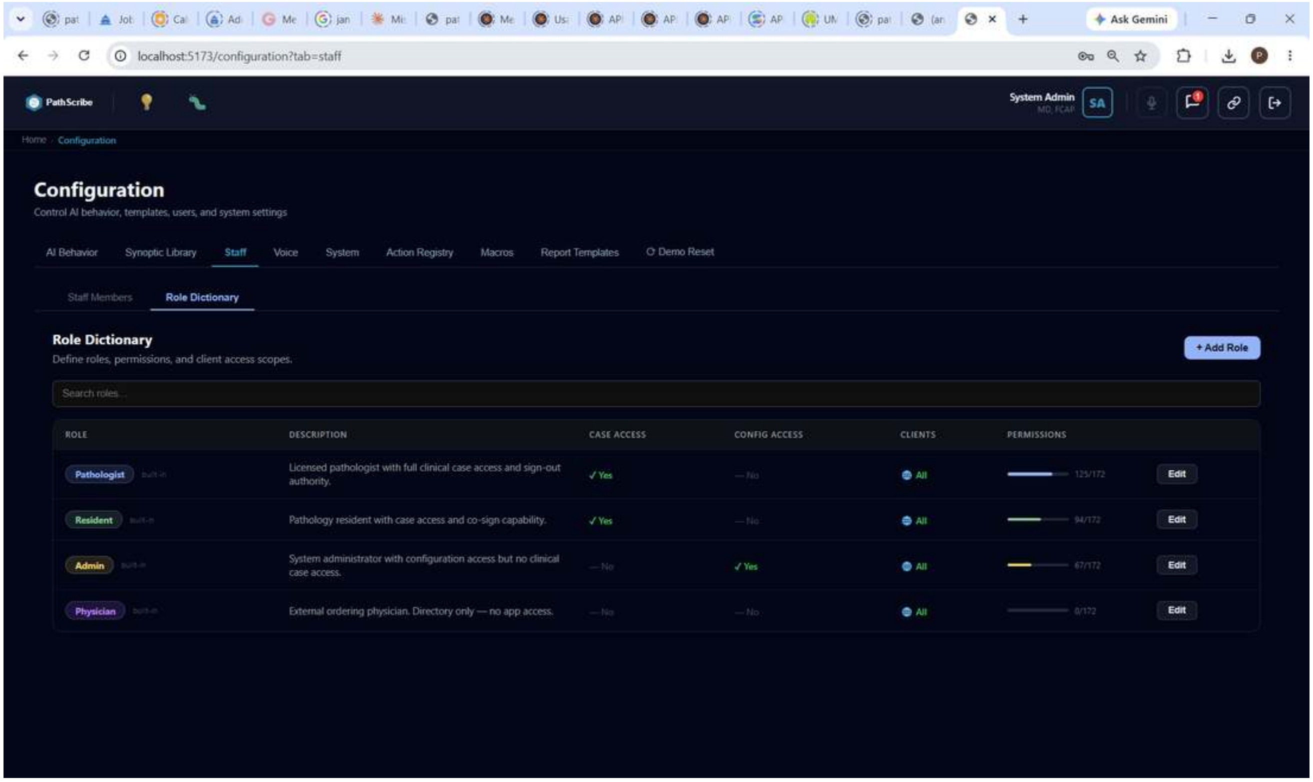
Field	Required	Notes
First & Last Name	Yes	Used on reports and result notifications.
NPI Number	Recommended	Used for billing validation and deduplication.
Fax / Email	At least one	Required for result delivery.
Preferred Contact	No	Drives automatic routing: Email, Fax, or Portal.
Client Affiliations	No	Links physician to one or more submitting facilities.
Status	Yes	Active / Inactive / Unverified. Unverified physicians do not receive notifications.

> **IMPORTANT**
>
> Unverified physicians will not receive result notifications until an admin adds a contact method and sets the record to Active. Check the Physicians table regularly for the amber Unverified indicator.

4. Role Management

Navigate to Configuration -> Staff -> Roles.

4.1 Built-in Roles



Role Dictionary -- the four built-in roles, with live permission counts out of 172 total actions.

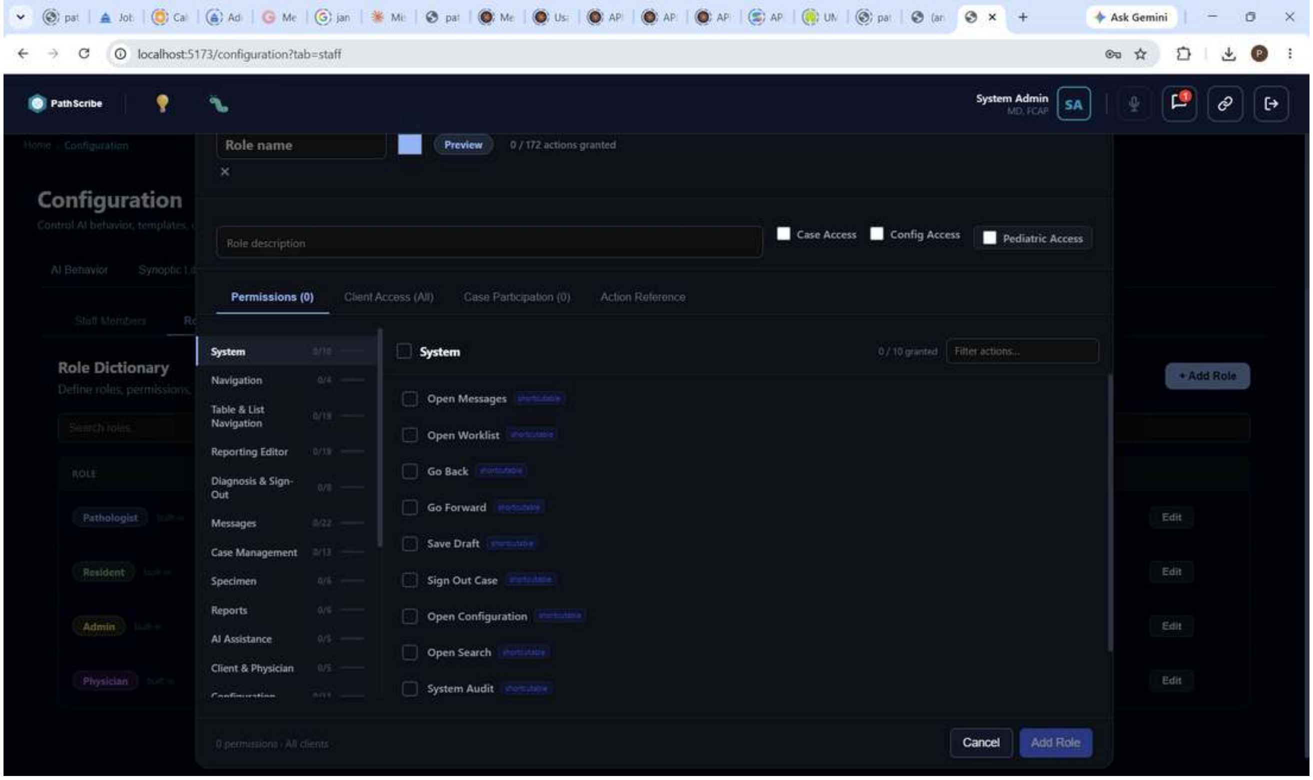
Role	Case access	Config access	Can be deleted	Description
Pathologist	Yes	No	No	Full clinical case access and sign-out authority. 125/172 permissions.
Resident	Yes	No	No	Case access with co-sign only. Sign-out gated by attending countersign. 94/172 permissions.
Admin	No	Yes	No	System configuration. No clinical access. 67/172 permissions.
Physician	No	No	No	External directory only. No app access. 0/172 permissions.

4.2 Custom Roles

- Click Add Role. Enter a name, description, and set Case Access and Configuration Access toggles.
- Custom roles can be fully edited and deleted. Built-in roles can only have their permissions adjusted.

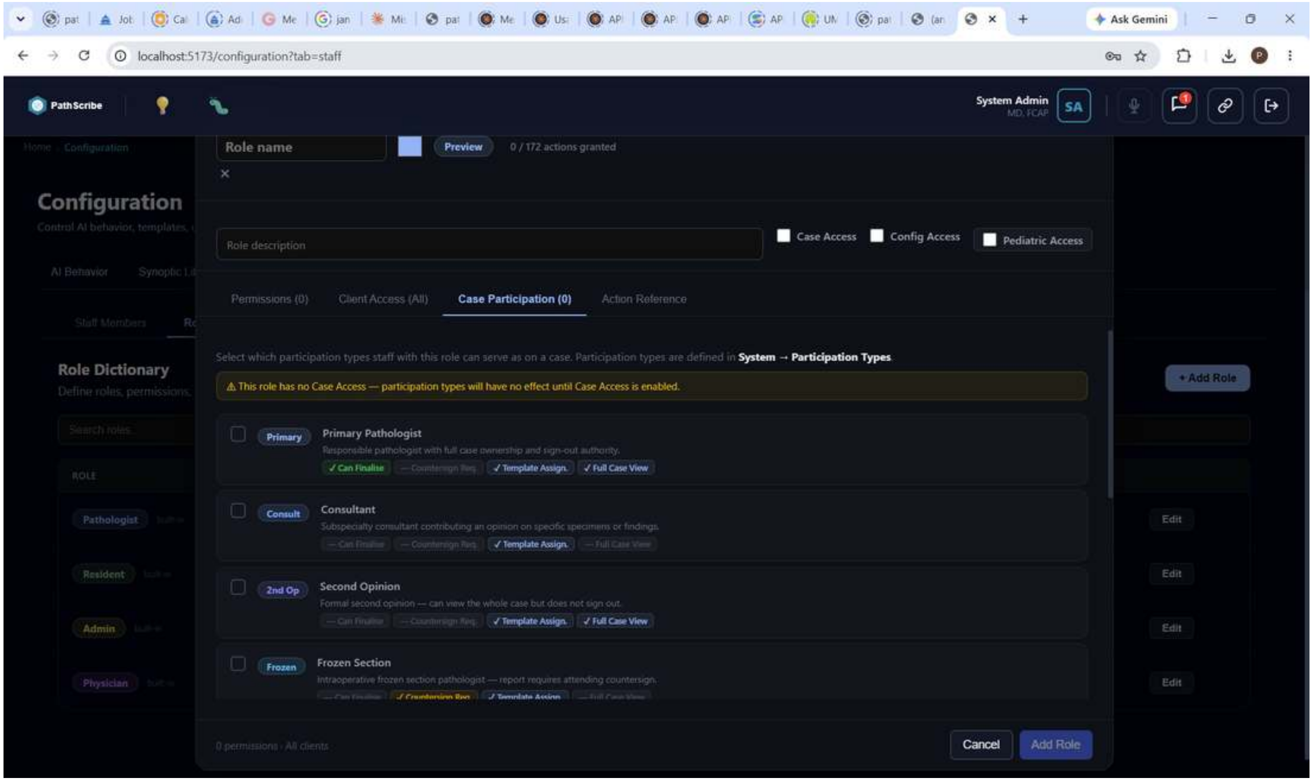
4.3 Permissions

Each role has a granular permissions list -- 172 individual actions across categories (System, Navigation, Table & List Navigation, Reporting Editor, Diagnosis & Sign-Out, Messages, Case Management, Specimen, Reports, AI Assistance, Client & Physician, Configuration, and more). The Add/Edit Role modal has four tabs: Permissions, Client Access, Case Participation, and Action Reference.



Add Role -- Permissions tab, grouped by category with a live granted-count.

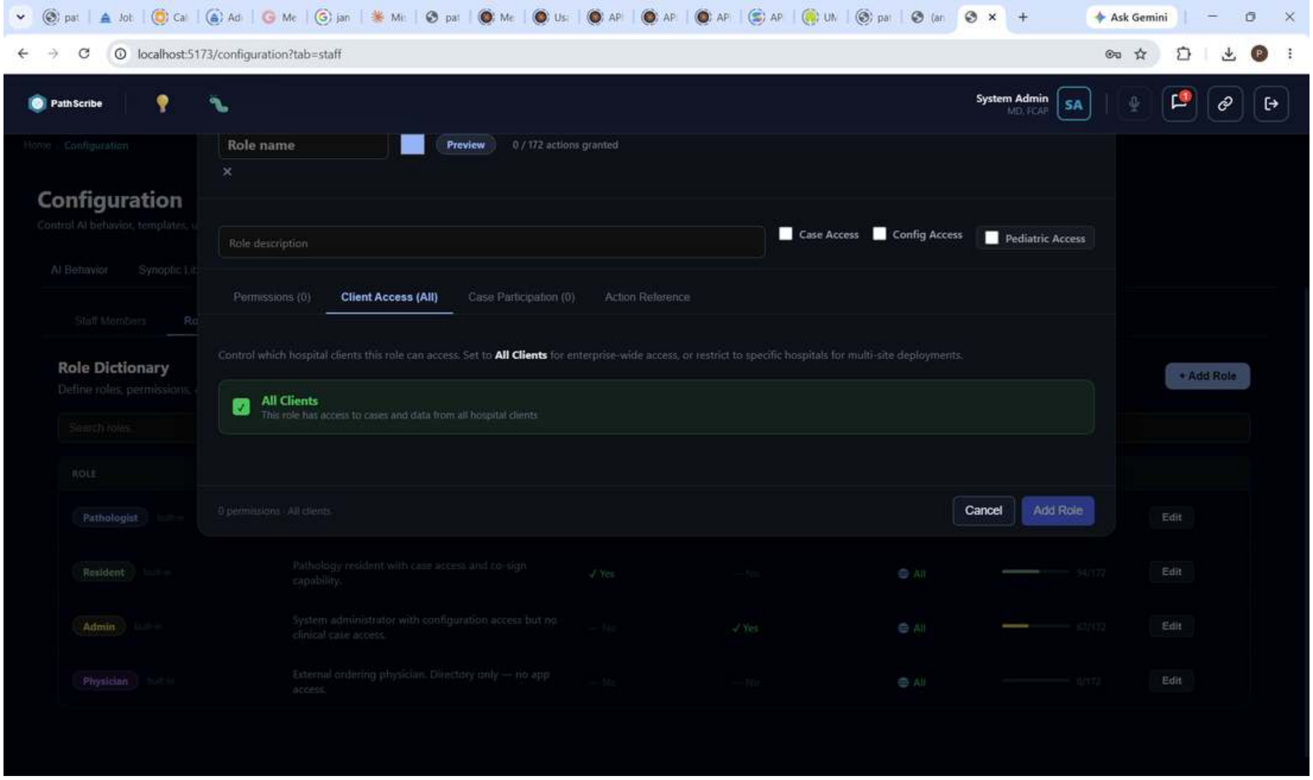
Case Participation determines which Participation Types (Section 18.5) staff with this role can serve as on a case -- e.g. Primary Pathologist, Consultant, Second Opinion, Frozen Section. Each type carries its own capability flags (Can Finalise, Countersign Required, Template Assignment, Full Case View).



Add Role -- Case Participation tab, linked to System -> Participation Types.

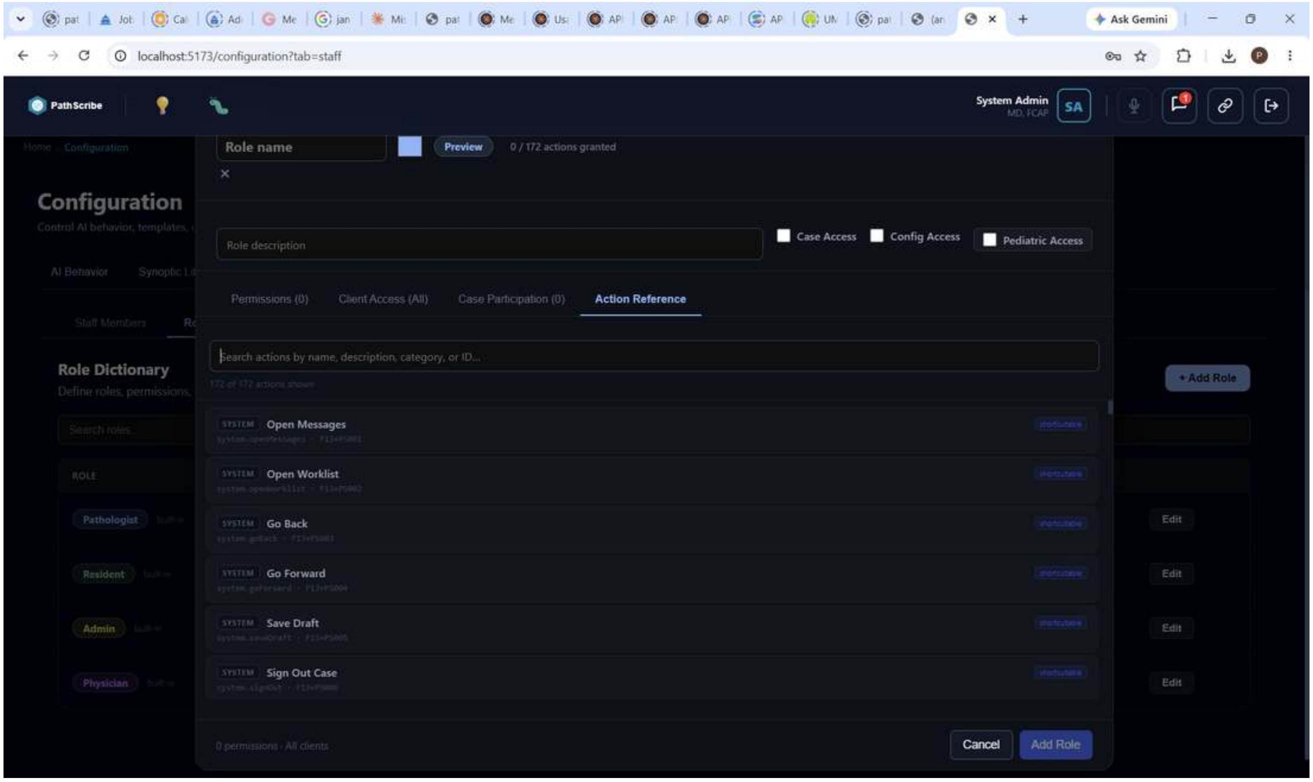
> **NOTE**
>
> A role with Case Access disabled will have no effective case participation regardless of which participation types are checked -- the two settings work together, not independently.

Client Access controls which hospital clients this role can see cases and data from -- All Clients for enterprise-wide access, or restricted to specific hospitals for multi-site deployments.



Add Role -- Client Access tab.

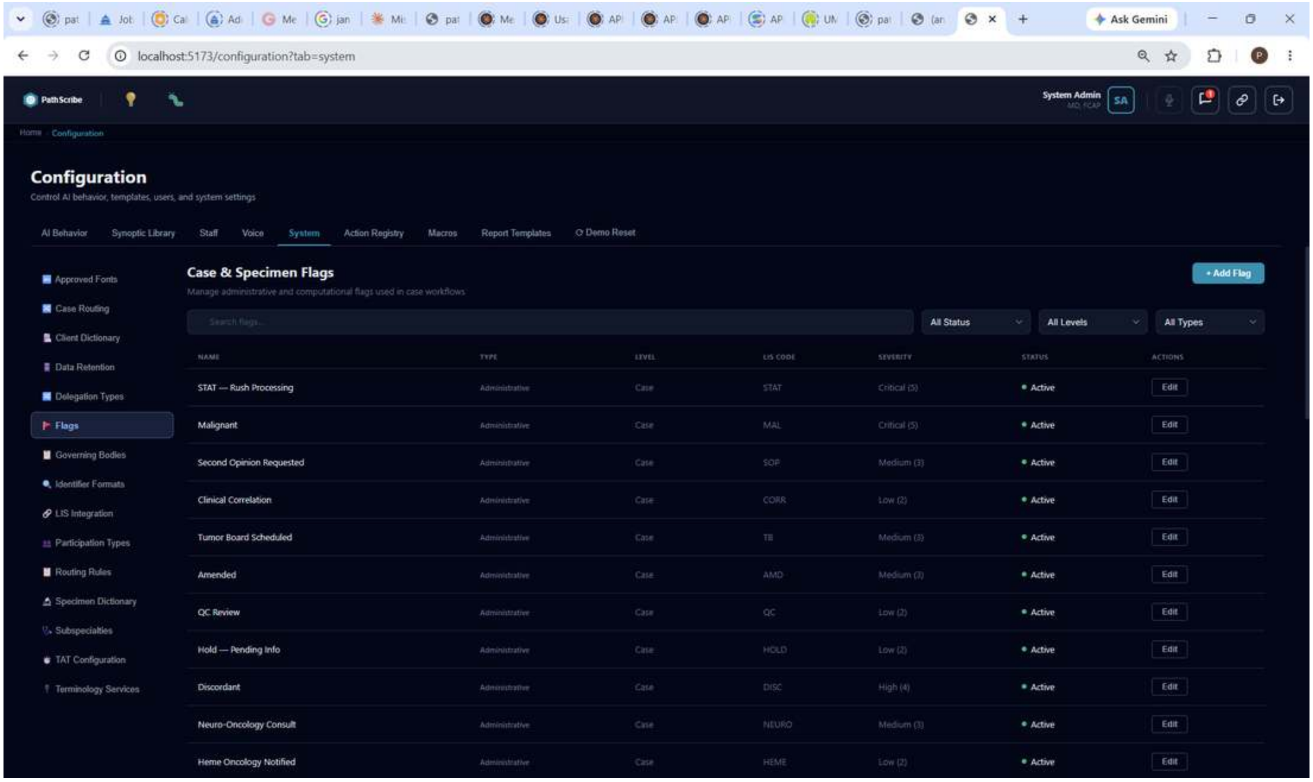
Action Reference is a searchable, read-only list of all 172 actions -- showing each action's internal ID and keyboard shortcut binding (e.g. system.openMessages -- F13+PS001) -- useful for confirming exactly what a permission checkbox actually grants.



Add Role -- Action Reference tab, with internal action IDs and shortcut bindings.

5. Flag Management

Navigate to Configuration -> System -> Flags. Flags are visual markers on cases and specimens that communicate clinical context or required actions.



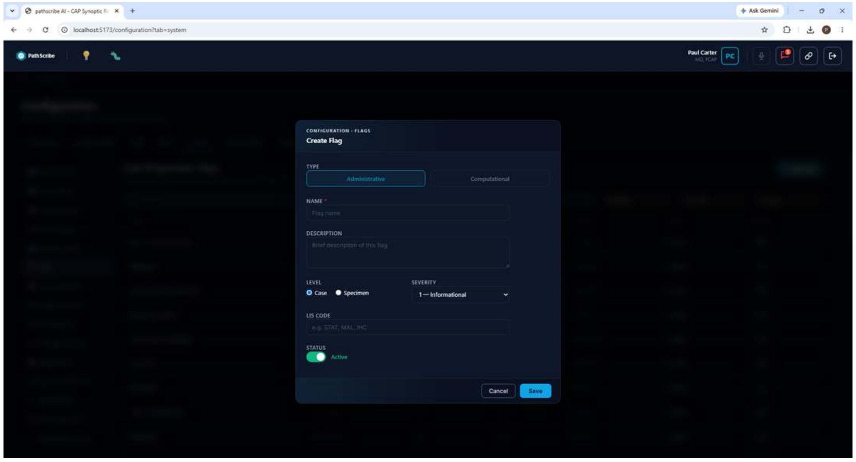
Case & Specimen Flags -- name, type, level, LIS code, severity, status.

5.1 Case Flags vs Specimen Flags

- Case flags apply to an entire case (e.g. "Tumor Board -- Thu 14:00", "QI Review Required", "Teaching Case").
- Specimen flags apply to a specific specimen within a case (e.g. "ER/PR/HER2 Pending", "BRAF Mutation", "Molecular Panel").

5.2 Creating a Flag

- Click + Add Flag. Choose Type: Administrative or Computational.
- Enter a name, description, Level (Case or Specimen), and Severity (1 -- Informational, up to 5 -- Critical).
- For specimen flags, enter the LIS Code -- used by your LIS to trigger this flag automatically on result receipt.



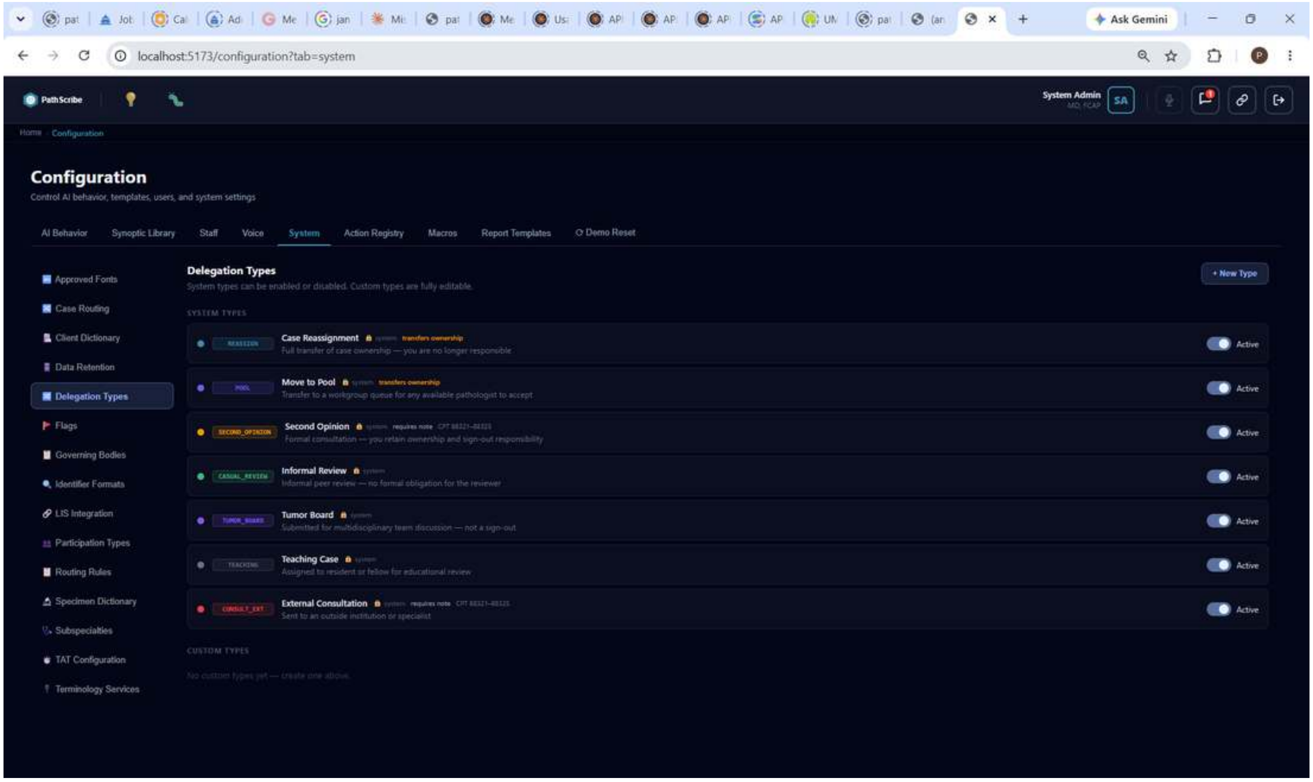
Create Flag -- Type, Level, Severity, LIS Code, Status.

5.3 Computational Flags

Data Source	Type	Description	Required fields
LIS Endpoint	Direct	REST call from PathScribe to the LIS result endpoint.	Source ID, API Endpoint URL.
AI Extraction	AI proxy	AI proxy parses unstructured LIS text to extract the result.	Extraction endpoint URL.
PathScribe Entry	Manual	Pathologist enters the result directly in the synoptic form.	None -- no endpoint required.

5.4 Delegation Types

Navigate to Configuration -> System -> Delegation Types. Seven built-in system types ship with PathScribe; system types can be enabled/disabled but not edited. Custom types are fully editable.



Delegation Types -- 7 system types, each with ownership-transfer and note/CPT requirements.

Type	Transfers Ownership?	Notes
Case Reassignment	Yes	Full transfer -- you are no longer responsible for the case.
Move to Pool	Yes	Transfer to a workgroup queue for any available pathologist to accept.
Second Opinion	No	Formal consultation; requires a note. CPT 88321-88325.
Informal Review	No	Informal peer review -- no formal obligation for the reviewer.
Tumor Board	No	Submitted for multidisciplinary team discussion -- not a sign-out.
Teaching Case	No	Assigned to a resident or fellow for educational review.
External Consultation	No	Sent to an outside institution or specialist; requires a note. CPT 88321-88325.

Click + New Type to create a Custom type: Label, Description, CPT Hint, accent colour, and Active toggle. The type immediately appears in the Delegate Modal for all active users.

6. AI Configuration

Navigate to Configuration -> AI Settings.

AI Behavior configuration screen.

Setting	Default	Description
AI Provider	Anthropic	Select Anthropic Claude or Google Gemini as the active AI provider.
Model	claude-sonnet-4-6	The specific model version. Claude Sonnet 4.6 recommended for clinical accuracy.
API Mode	Proxy	Proxy routes calls through the backend (production). Direct is development only.
Confidence Threshold	75%	Fields below this threshold show "AI: not found" instead of a suggestion. Controls which fields appear in the AI Triage modal at finalisation.
Gross-Driven AI	On	AI analyses the gross description and pre-populates synoptic fields automatically.
Microscopic-Driven AI	On	AI re-analyses after microscopic text is entered, refining suggestions and evaluating protocol assignments (Section 9.2, Phase 3).
Auto-insert suggestions	Off	If on, AI suggestions are applied to fields without requiring pathologist confirmation.
Voice AI	Gemini active	Enable Gemini-powered voice command recognition.

CORRECTED -- June 2026

The Model default below was previously listed as "claude-sonnet-4-20250514" -- a stale model ID found hardcoded in .env and corrected to claude-sonnet-4-6. If your deployment still references the old ID anywhere, update it.

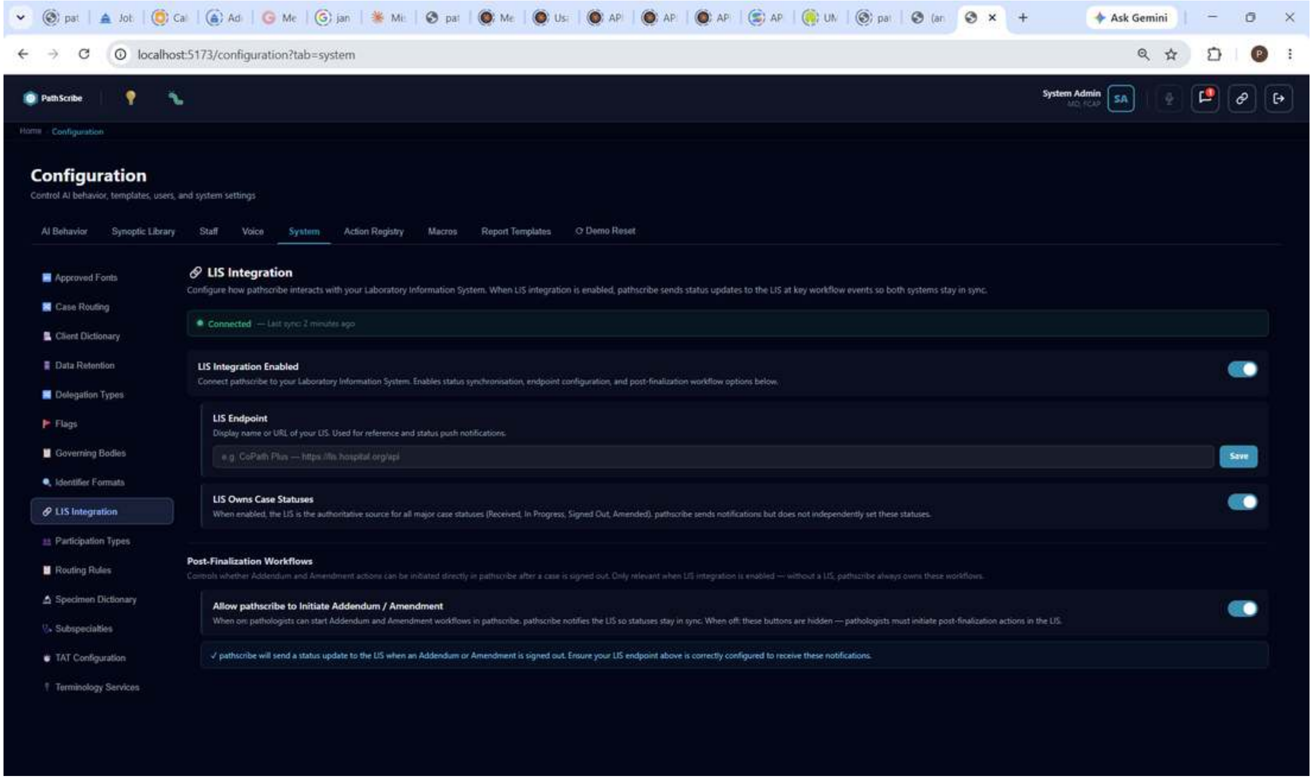
IMPORTANT

Changes to AI settings take effect for new sessions. Active users must log out and back in to see updated AI behaviour.

7. System Configuration

Navigate to Configuration -> System. The left sidebar gives access to all system-wide settings.

7.1 LIS Integration



LIS Integration -- connection status, endpoint, and post-finalization workflow controls.

Setting	Default	Description
LIS Integration Enabled	Off	Master switch. Enable only when the HL7/FHIR endpoint is configured and tested.
LIS Endpoint	(empty)	Display name or URL of your LIS. Used for reference and status push notifications.
LIS Owns Case Statuses	On	The LIS is authoritative for all major case statuses (Received, In Progress, Signed Out, Amended). PathScribe sends notifications but does not independently set these statuses.
Allow PathScribe to Initiate Addendum / Amendment	On	Lets pathologists start Addendum/Amendment workflows directly in PathScribe, which then notifies the LIS to keep statuses in sync. Only relevant when LIS Integration is enabled -- without a LIS, PathScribe always owns these workflows.

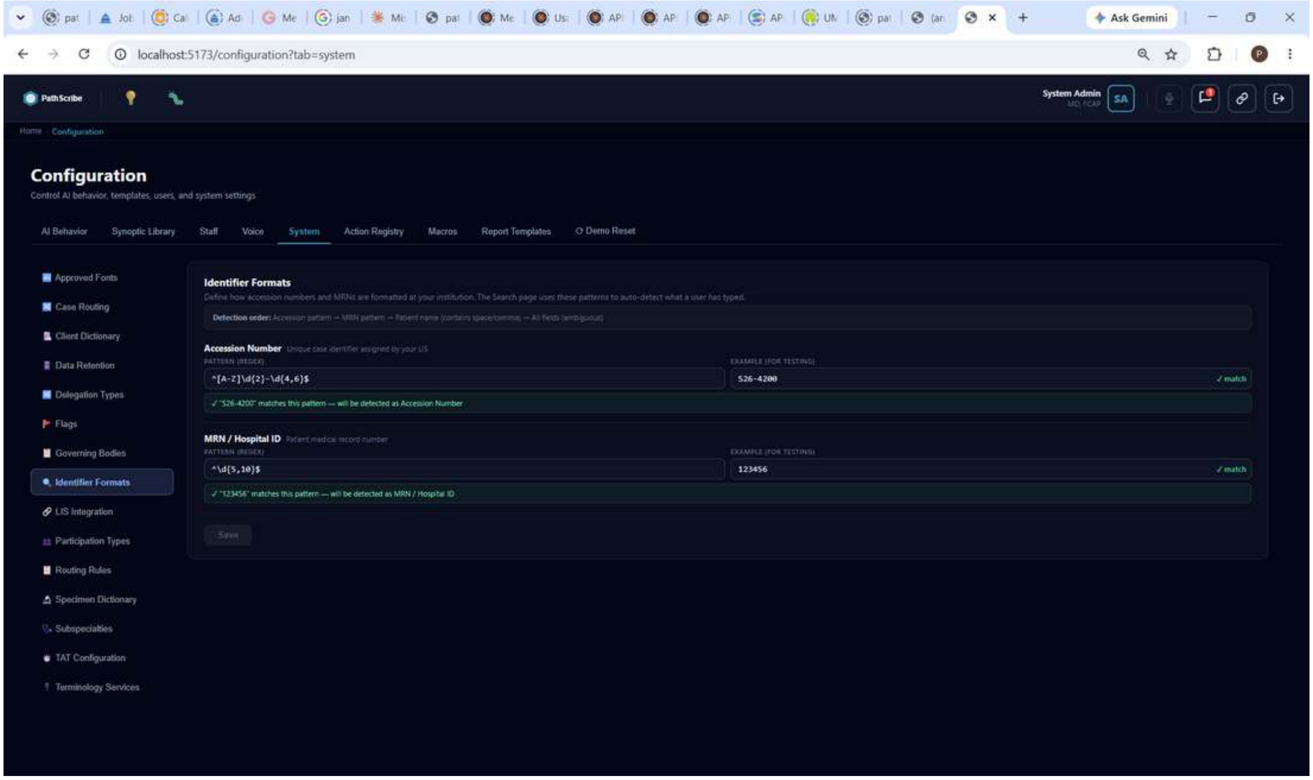
7.2 Jurisdiction

Jurisdiction	SNOMED release	ICD variant
US	SNOMED CT US Edition	ICD-10-CM
CA (Canada)	SNOMED CT CA Edition	ICD-10-CA
GB_EW (England & Wales)	SNOMED CT UK Edition	ICD-10 (WHO)

Jurisdiction	SNOMED release	ICD variant
GB_SCT (Scotland)	SNOMED CT UK Edition	ICD-10 (WHO)
IE (Ireland)	SNOMED CT IE Edition	ICD-10 (WHO)

7.3 Identifier Formats

- Go to Configuration -> System -> Identifier Formats.
- Enter the regex pattern for your accession number format. The Example field lets you test it live.
- Enter the regex pattern for your MRN format and test similarly. Click Save.

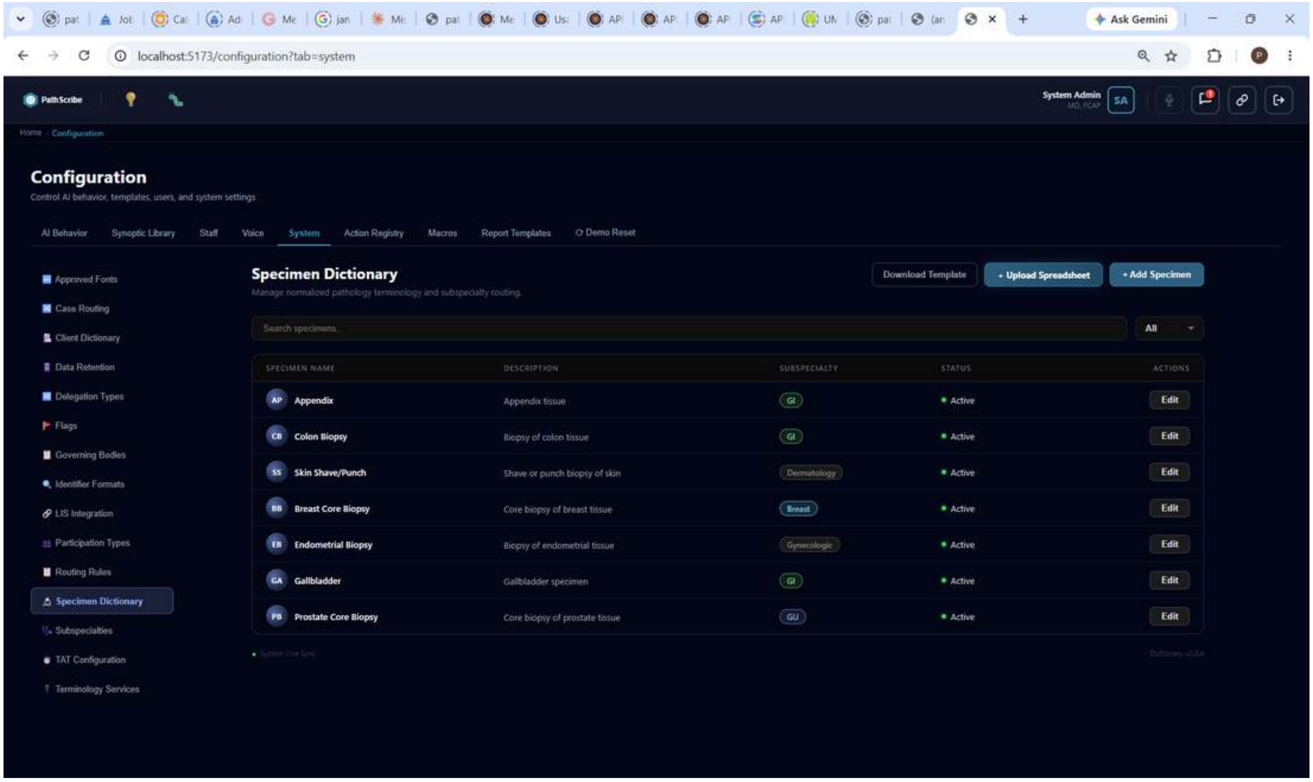


Identifier Formats -- live regex pattern test for Accession Number and MRN/Hospital ID.

> **NOTE**
>
> Detection order on the Search page: Accession pattern, then MRN pattern, then patient name (contains a space or comma), then all fields if still ambiguous.

7.4 Specimens

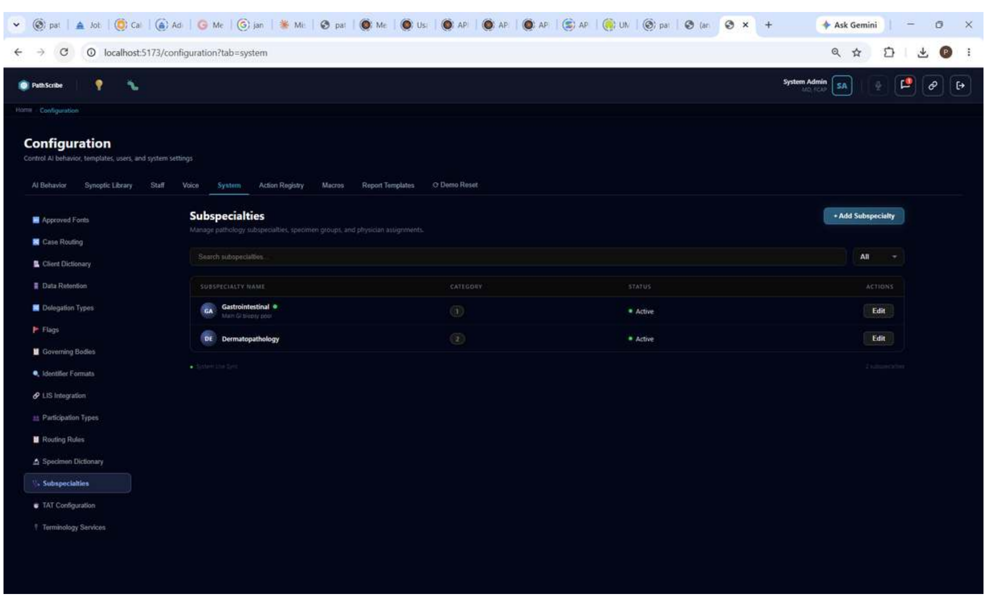
The specimen dictionary drives autocomplete in Case Search and case creation, and maps each specimen type to a subspecialty. Add, edit, or deactivate specimen types here, or upload a spreadsheet of many at once.



Specimen Dictionary -- specimen name, description, subspecialty, and status.

7.5 Subspecialties

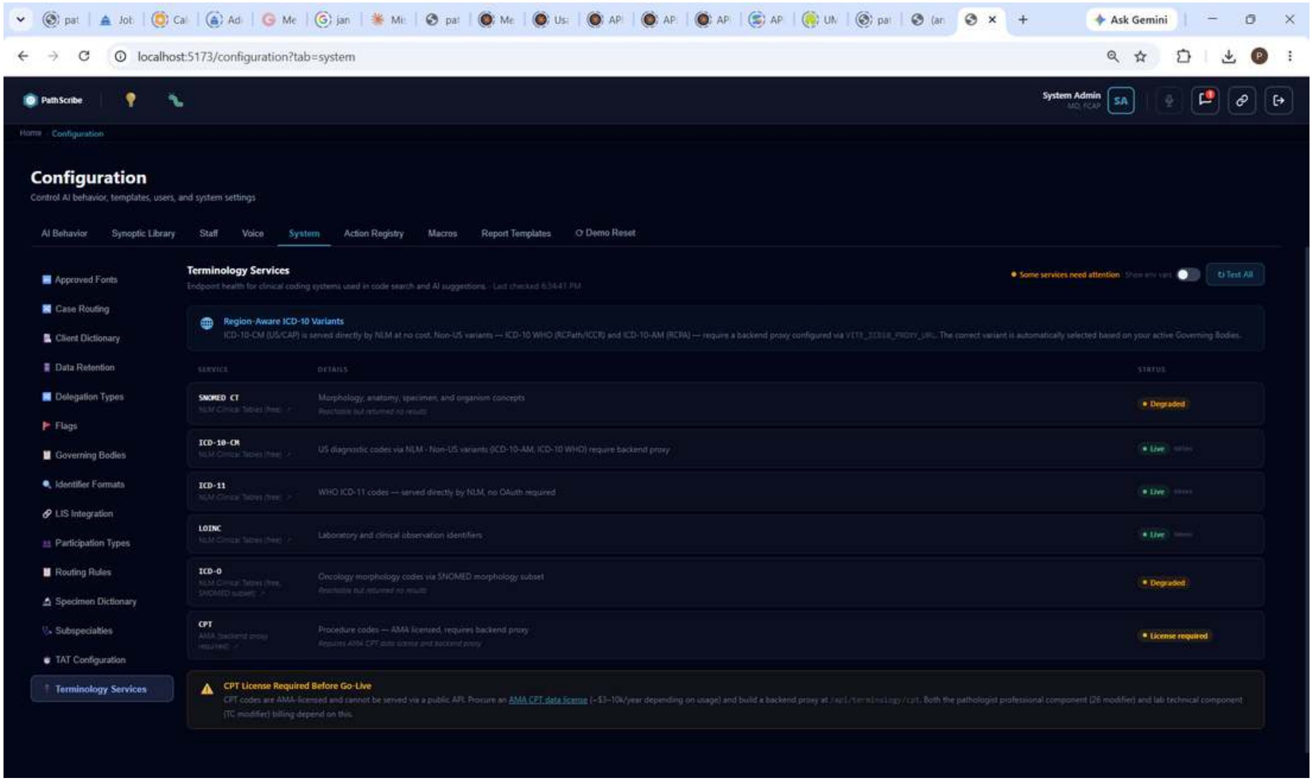
Manage pathology subspecialties, the specimen groups that map to them, and physician assignments. Define subspecialty pools for the delegation workflow -- cases can be routed to a subspecialty pool rather than an individual pathologist.



Subspecialties -- e.g. Gastrointestinal ("Main GI biopsy pool"), Dermatopathology.

7.6 Terminology Services

Navigate to Configuration -> System -> Terminology Services. Each service shows a status badge (Live / Degraded / License Required) and last-checked timestamp. A green/Live badge is required before go-live.



Terminology Services -- SNOMED CT, ICD-10-CM, ICD-11, LOINC, ICD-O, and CPT, with live status.

> **CPT LICENSE REQUIRED BEFORE GO-LIVE**
>
> CPT codes are AMA-licensed and cannot be served via a public API. A CPT data licence and a backend proxy are both required before CPT codes can be displayed -- matching Pre-Launch Checklist item 5/6 (Section 14).

8. Report Template Builder

Key architecture: A Part is an independently-versioned module (header, body section, footer). A Template is an ordered assembly of Parts. Updating a Part instantly updates every template that references it. PathScribe ships with 14 pre-built CAP-compliant Parts.

> **COUNT CONFIRMED**
>
> Confirmed directly from application code and from two independent screenshots of the New Template sidebar ("BODY PARTS 10"): the current count is 14 built-in Parts (4 header/footer + 10 body), listed in full in Part II, Appendix A. One older Part Library screenshot elsewhere showed "BODIES 9," which appears to predate the 10th body Part being added rather than reflect the current count.

SEE PART II

This section gives a brief orientation only. Full operational detail -- every component type, the Inspector panel, AI Generation configuration, Template Assembly, Routing Rules, and step-by-step workflows -- is covered in depth in Part II of this guide: Report Template Configuration.

8.1 Accessing the Template Builder

Navigate to Configuration -> Template Builder. The builder has two tabs: Part Library and Templates.

8.2 -- 8.9

Creating a Part, the Canvas and Palette, Canvas Interactions, the Inspector Panel, AI Generation Configuration, Previewing a Part, Assembling a Template, and Publishing a Template are all covered in full in Part II, Sections 4-5.

8.10 How Templates Drive Report Output

FIXED -- June 2026

Published templates now directly drive both the Report Preview pane and the final printed PDF -- reordering Parts, label formatting (position/transform/weight/decoration/font size), and Visibility Conditions configured here will appear in generated reports immediately. Previously, generated reports used a separate hardcoded narrative configuration disconnected from this builder; admins testing template changes here may have seen no effect on real output. That gap is closed -- see Part II, Section 7 for the corrected pipeline detail.

9. Synoptic Report Lifecycle

A synoptic report in PathScribe is a structured, field-by-field clinical summary attached to a specific specimen and template. One case can have many synoptic reports -- one per specimen per protocol.

9.1 Template-to-Specimen Assignment

- AI-driven: PathScribe proposes the most appropriate published template based on specimen description, clinical indication, and gross description. The pathologist confirms, modifies, or overrides.
- Manual: The pathologist selects a template via Add Synoptic in the right panel.

NOTE

Templates must be published in the Template Builder (Section 8 / Part II) before they appear in the assignment list.

9.2 The Seven Phases

Phase 1 -- Case Creation & Specimen Registration. Case arrives from LIS (Co-Pilot) or manual entry (Standalone). Each specimen is registered with a label, type, and description.

Event	Audit Record
Case created / received from LIS	CASE_CREATED -- timestamp, source, accession number
Specimen registered	SPECIMEN_REGISTERED -- label, description, case ID

Phase 2 -- Gross Submission & AI Protocol Assignment. AI analyses the gross description and proposes a synoptic template per specimen. The pathologist accepts, modifies, or rejects.

Event	Audit Record
Gross description saved	GROSS_SAVED -- content hash, word count, timestamp
AI template proposal	AI_PROTOCOL_PROPOSED -- templateId, templateName, confidence
Template assigned	SYNOPTIC_ASSIGNED -- instanceId, templateId, specimenId, assignedBy

Phase 3 -- Microscopic Entry & Protocol Re-Evaluation. When the microscopic description is saved, the AI re-evaluates whether the assigned template is still appropriate. If findings suggest a different protocol, a Protocol Change Review is raised.

NEW in v0.9.0

The Protocol Change Review modal lists each proposed change with the current template, proposed replacement, AI reasoning, and confidence score. The pathologist selects which changes to apply and commits. Rejected changes are logged but not applied.

Event	Audit Record
Microscopic saved	MICRO_SAVED -- content hash, timestamp
Protocol change proposed	PROTOCOL_CHANGE_PROPOSED -- specimenId, currentTemplateId, proposedTemplateId, confidence
Protocol change accepted/rejected	PROTOCOL_CHANGE_ACCEPTED / REJECTED -- instanceId, pathologistId

Phase 4 -- Synoptic Field Completion. AI Copilot pre-populates fields with confidence badges (green >= 85%, amber 50-84%, red < 50%). Fields can be accepted with one click or overridden. Deferred synoptics (awaiting ancillary results) are excluded from the current sign-out and trigger an addendum workflow when completed.

Phase 5 -- Pre-Finalisation Review. When the pathologist clicks Finalise, PathScribe opens the Pre-Finalisation Review -- a full-screen two-pane modal -- before collecting credentials.

NEW in v0.9.0

Left pane: all specimens and synoptic reports in transmission order, drag to reorder, exclude individual synoptics from this transmission. Right pane: live preview of the complete report output as it will be transmitted to the LIS, updating in real time. Signing panel: biometric (if enrolled) or password -- this constitutes the digital authorisation.

Phase 6 -- LIS Transmission (Co-Pilot Mode). PathScribe transmits the finalised synoptic content to the LIS in the transmission order set in the Pre-Finalisation Review. Excluded synoptics remain as drafts for addendum. In Standalone mode, there is no outbound LIS transmission.

Phase 7 -- Post-Finalisation: Amendments & Addenda. Amendment: a corrective change to a finalised report, transmitted to the LIS as a corrected report. Addendum: an official addition to a finalised report, used for deferred synoptics and supplementary information.

9.3 Multi-Specimen Cases

Each specimen has its own independent synoptic report instance. Template assignment, AI suggestions, and field completion operate independently per specimen. The Pre-Finalisation Review shows all specimens together for ordering and exclusion before signing.

9.4 Resident Counter-Sign Workflow

When a Resident submits for sign-out, PathScribe routes it to their assigned attending for countersign rather than directly finalising. The attending countersigns (case proceeds to Pre-Finalisation Review) or returns it with a comment.

9.5 Audit Trail Summary

Every interaction generates an immutable, append-only audit record. Key properties: timestamp (UTC, millisecond precision), user ID and session ID, event type, affected entity IDs, content hashes, and AI metadata where applicable. Accessible at Home -> System Audit. Export to CSV for compliance reporting.

> **NOTE**
>
> The audit log is designed to satisfy CAP, RCPath, CLIA, and ISO 15189 audit requirements. Content hashes allow verification that stored reports have not been altered without exposing PHI.

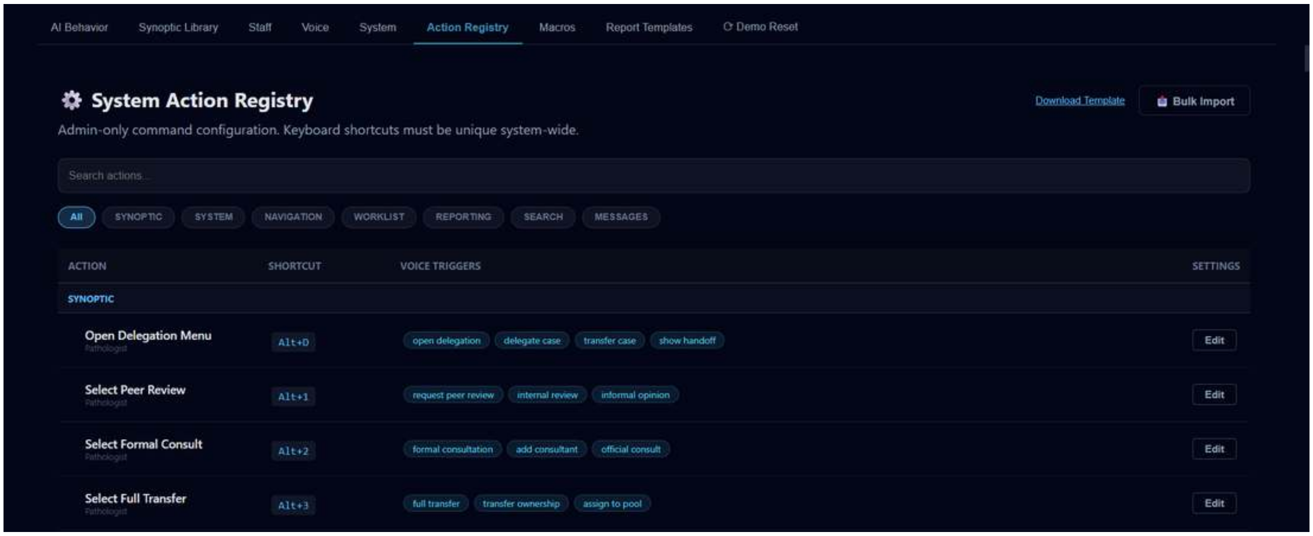
10. Action Registry & Voice Control

PathScribe has 139 registered actions -- every action available via mouse or keyboard is also available by voice. The Action Registry is at Configuration -> Action Registry.

10.1 Architecture

Layer	Function
Voice Provider	Captures microphone input via the Web Speech API. Normalises transcripts.
Action Registry	Matches transcripts against registered voice triggers using exact match, learned triggers, then fuzzy scoring.
Event Dispatcher	Fires a PATHSCRIBE_ custom DOM event. Handlers respond per-page and per-modal.
Learning Layer	Records unrecognised phrases. When the user confirms what they meant, the mapping is stored permanently.

10.2 Coverage

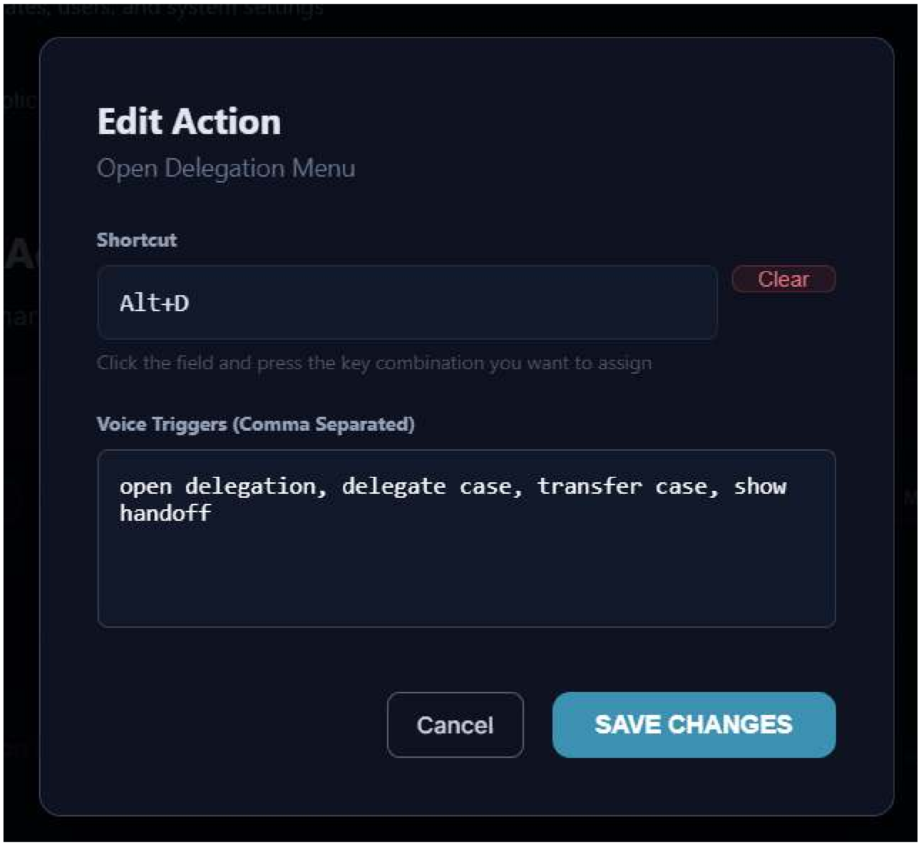


Action Registry -- full list of registered actions with keyboard shortcuts and voice triggers.

Area	Actions
System, Navigation, Messages (AppShell)	44 actions
Worklist	30 actions
Synoptic Report and sub-components	60 actions
Search	3 actions
Configuration tab navigation	2 actions
Total	139 actions / 60 with keyboard shortcuts

10.3 Editing an Action

- Navigate to Configuration -> Action Registry.
- Find the action by name or ID. Click to expand.
- Edit the keyboard shortcut or voice triggers (comma-separated phrases). Click Save -- changes apply immediately.



Editing an action -- keyboard shortcut and voice trigger phrases.

Persistence: edits persist across deployments. New actions added by developers are merged automatically. To reset to defaults, clear the ps_action_registry key from localStorage.

11. Audit Log

Navigate to Home -> System Audit. The audit log records all significant system events -- immutable and append-only.

- User login and logout events.
- Case creation, status changes, finalisation, and amendments.
- AI suggestion accept and reject events (confidence scores recorded).
- User account create, edit, and deactivate.
- Configuration changes (all settings, flag, role, template edits).
- Delegation and pool claim events.
- Computational test orders, cancellations, and result receipt.

Filter by date range, user, or event type. Click Export CSV for compliance reporting.

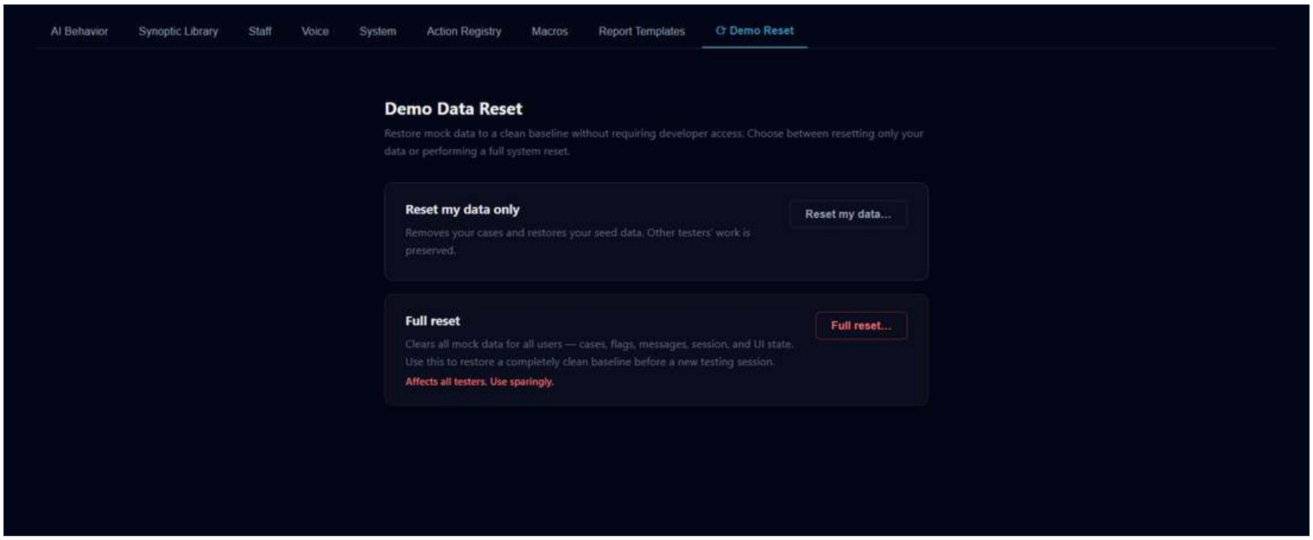
12. System Information

Click your initials badge -> System Information to view: application version and build date, AI provider and model, API connectivity status, and browser/OS details for support diagnostics. Click Copy to Clipboard to generate a formatted support report. The copy redacts the user's name and ID -- safe to share with PathScribe support.

13. Demo Reset (Development / Staging Only)

Navigate to Configuration -> Demo Reset. Visible in development and staging environments only.

- Reset my data only -- restores your hospital's cases without affecting other testers.
- Full reset -- restores all data for all users across all hospitals. Use sparingly.



Demo Reset -- per-user and full reset options.

14. Pre-Launch Implementation Checklist

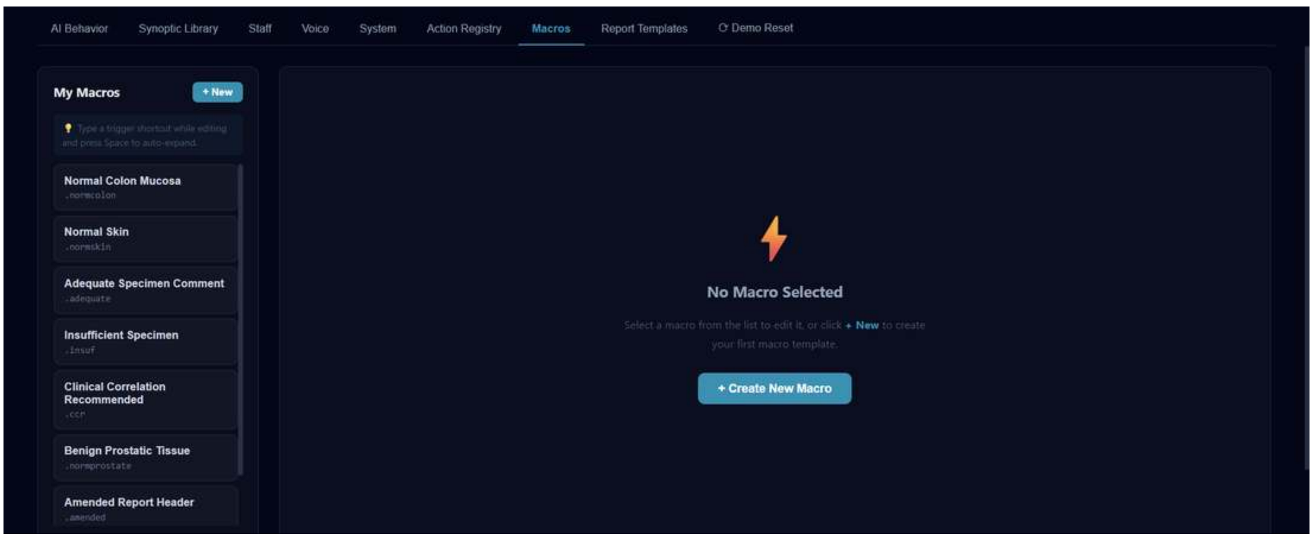
All items below must be completed before a customer goes live. Do not grant production access until every item is verified.

#	Requirement	Owner
1	Anthropic API key stored in backend secrets manager (never in VITE_ env vars)	Engineering
2	Business Associate Agreement (BAA) signed with Anthropic before any real PHI is processed	Legal
3	Anthropic training data opt-out confirmed for API account	Engineering
4	Production backend proxy built for all AI calls -- Vite dev proxy is development only	Engineering
5	AMA CPT data licence procured (approx. $3-10k/year) before CPT codes are displayed	Legal
6	CPT backend proxy endpoint built at /api/terminology/cpt	Engineering
7	UMLS licence obtained for full SNOMED CT	Legal
8	SNOMED CT production endpoint configured via UMLS API with backend key injection	Engineering
9	Non-US ICD-10 variant proxy configured if customer uses RCPath, RCPA, or ICCR protocols	Engineering
10	Firestore encryption at rest confirmed enabled in GCP project	Engineering
11	Flag definitions seeded for customer jurisdiction and lab type	Implementation

#	Requirement	Owner
12	Governing body configured for customer region (CAP / RCPath / RCPA / ICCR)	Implementation
13	AI provider set to Claude Sonnet 4.6 (recommended for clinical accuracy)	Implementation
14	All Terminology Services health checks passing	Implementation
15	PHI de-identification implemented before AI prompt submission with real patient data	Engineering
16	Penetration test completed by external vendor before go-live	Security
17	Report PDF generation Cloud Function (functions/main.py, ReportLab) deployed on Blaze plan; VITE_REPORT_PDF_ENDPOINT configured; CORS origin allowlist matches production frontend URL	Engineering

15. Macros

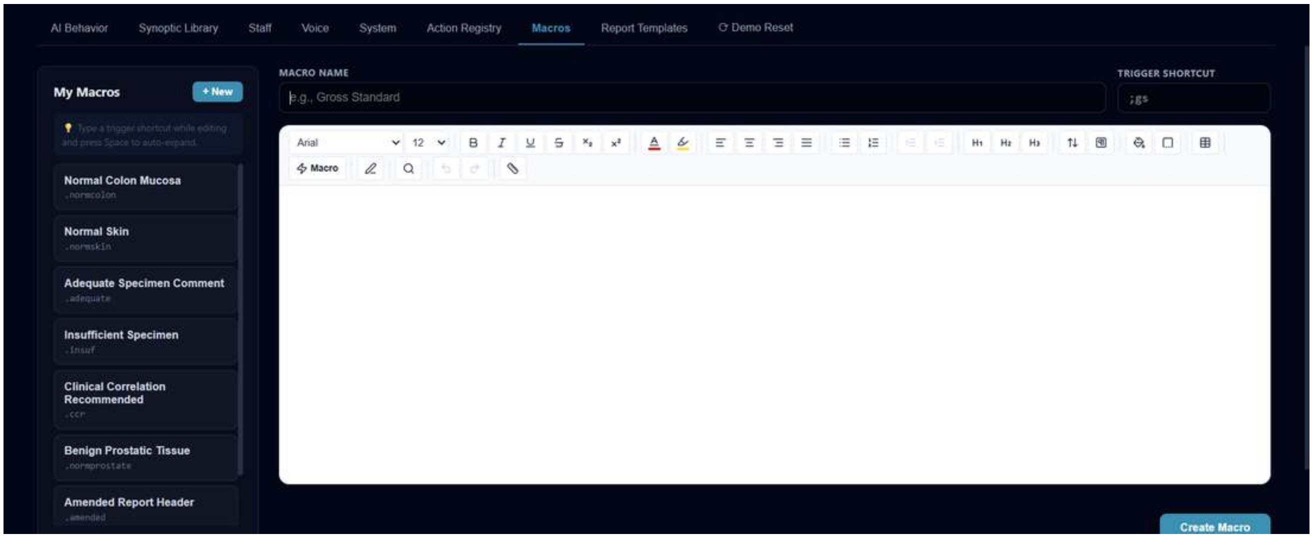
Navigate to Configuration -> Macros. Macros are reusable text snippets a pathologist can insert into Gross Description, Microscopic Description, or other free-text fields by typing a short trigger shortcut.



My Macros -- saved macro list with trigger shortcuts.

15.1 Creating a Macro

1. Click + New (or + Create New Macro from the empty state).
2. Enter a Macro Name and a Trigger Shortcut (e.g. .normcolon).
3. Compose the macro content in the rich text editor -- formatting, headings, lists, and tables are all supported, the same as the main diagnostic text fields.
4. Click Create Macro.



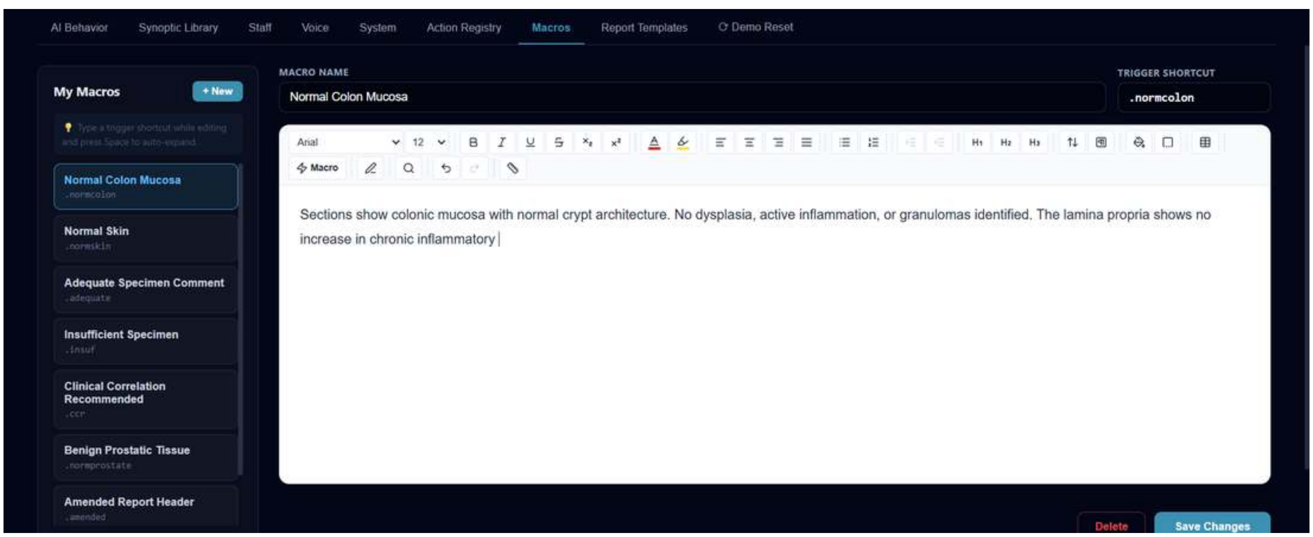
New Macro -- name, trigger shortcut, and full rich-text editor.

15.2 Using a Macro

Type the trigger shortcut while editing any supported text field, then press Space to auto-expand it into the full macro content.

15.3 Editing a Macro

Click any macro in the list to open it in the same editor used to create it. Changes apply the next time the trigger is used.



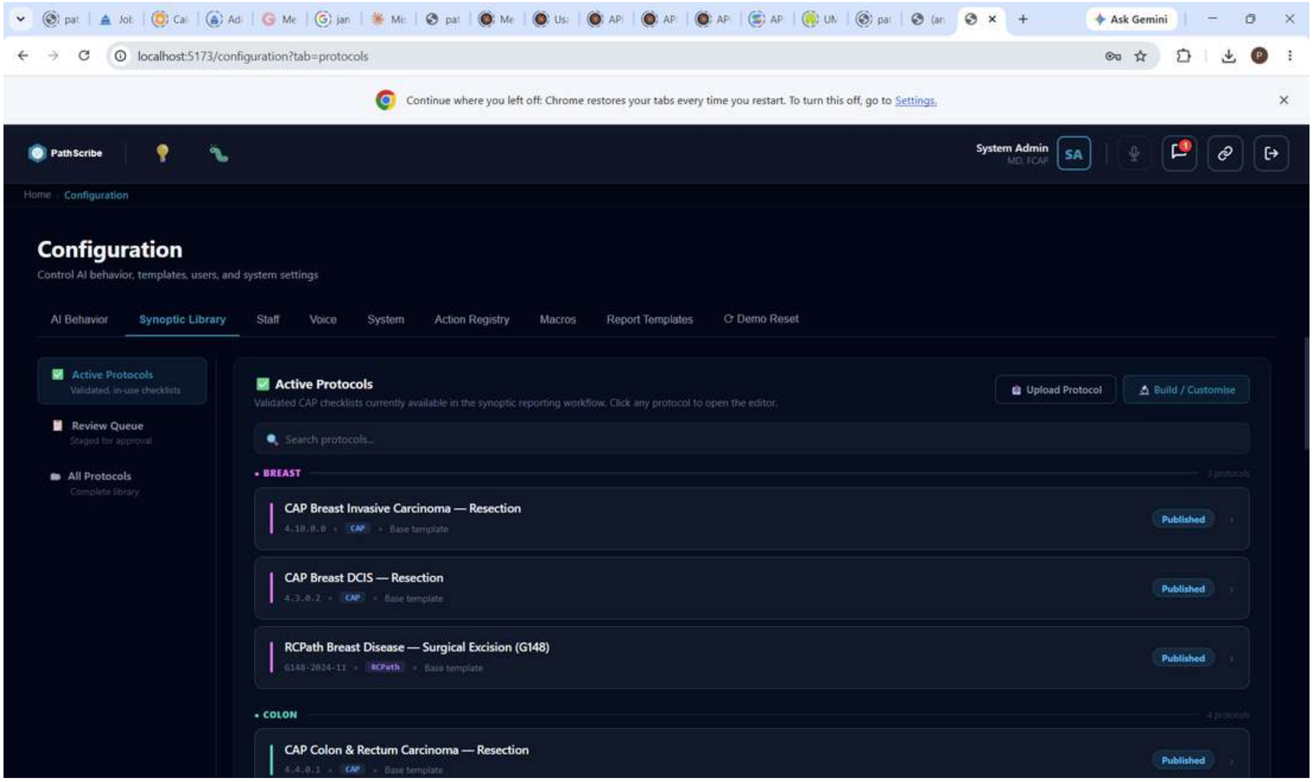
Editing an existing macro -- full content shown, with Delete and Save Changes.

16. Synoptic Library

Navigate to Configuration -> Synoptic Library. This is where CAP/RCPath/custom synoptic checklists themselves are managed -- distinct from Report Templates (Section 2.4 / Part II). Three views, in the left sidebar: Active Protocols, Review Queue, and All Protocols.

16.1 Active Protocols

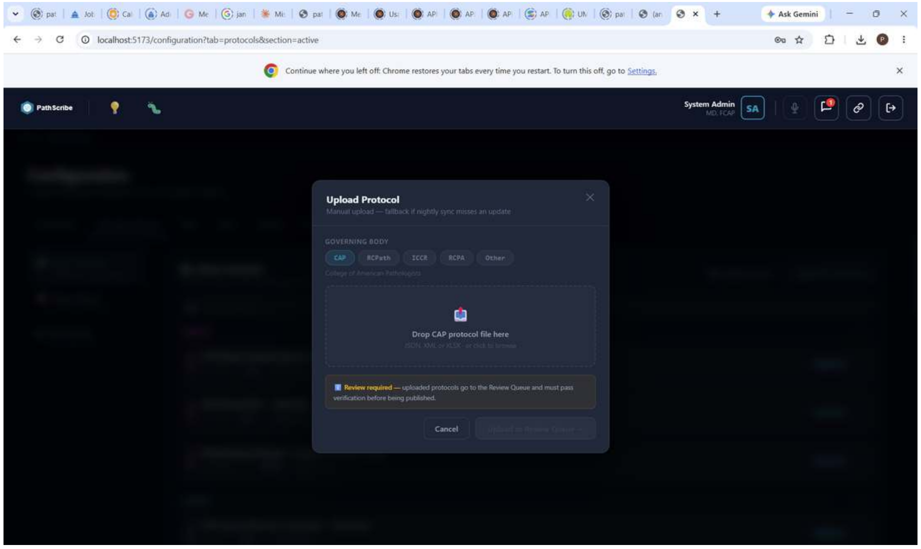
Validated, in-use checklists currently available in the synoptic reporting workflow, grouped by subspecialty. Each entry shows its version number, source (CAP / RCPath), and whether it's a Base template. Click any protocol to open it in the editor.



Active Protocols -- grouped by subspecialty, with version and source badges.

16.2 Upload Protocol

Manual upload is the fallback path for when the nightly sync misses an update. Select a governing body (CAP, RCPath, ICCR, RCPA, or Other) and drop a protocol file (JSON, XML, or XLSX).

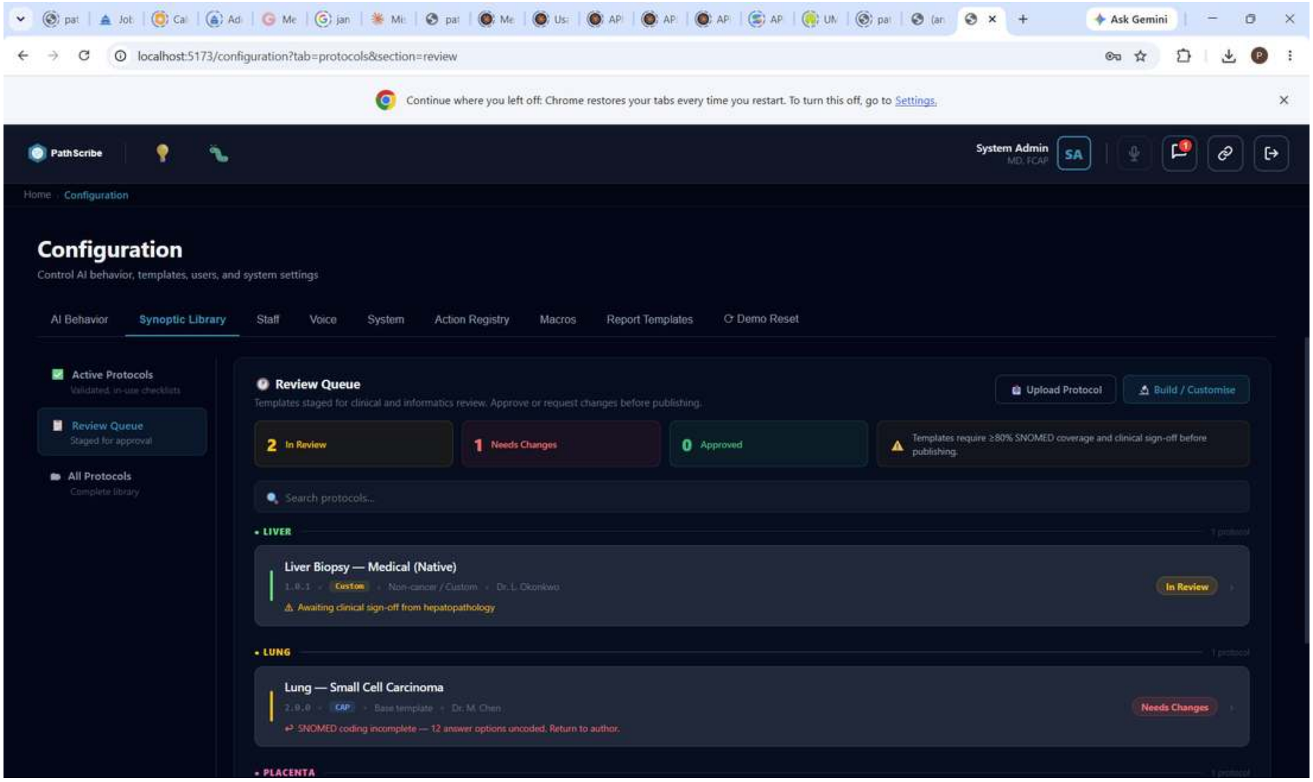


Upload Protocol -- governing body selection and file drop.

> **NOTE**
>
> Uploaded protocols always go to the Review Queue and must pass verification before being published -- there is no way to skip review on a manually uploaded protocol.

16.3 Review Queue

Templates staged for clinical and informatics review before publishing, with three status counters: In Review, Needs Changes, Approved. Custom/non-cancer protocols can be submitted by an individual pathologist, not just sourced from CAP/RCPath.

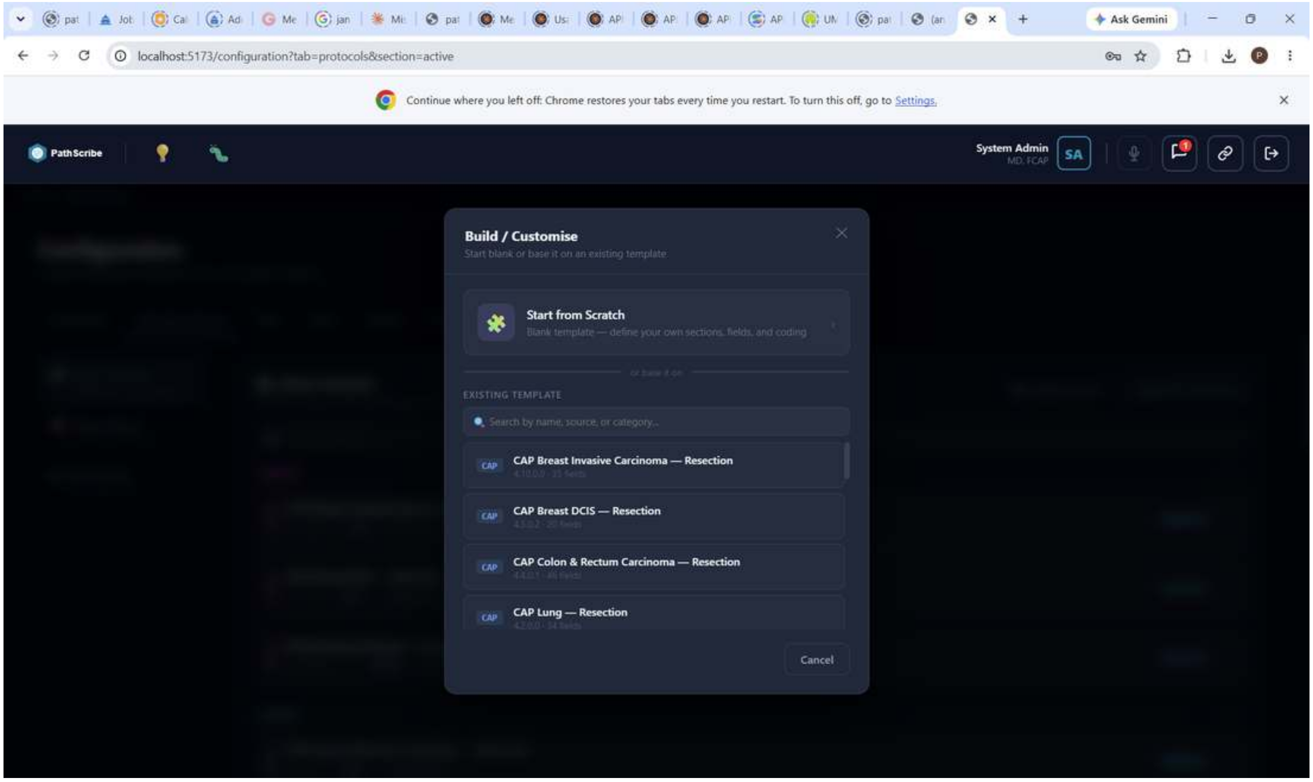


Review Queue -- status counters and per-protocol review notes (e.g. incomplete SNOMED coding).

> **KNOWN LIMITATION (as of this writing)**
>
> Publishing requires at least 80% SNOMED coverage and clinical sign-off. A protocol flagged "Needs Changes" (e.g. for incomplete SNOMED coding) must be returned to its author and resubmitted -- it cannot be force-published from this screen.

16.4 Build / Customise

Opens the synoptic form editor for building a new protocol from scratch or customising an existing one -- the field-level checklist editor, distinct from the Report Template Part Builder (Part II, Section 4).



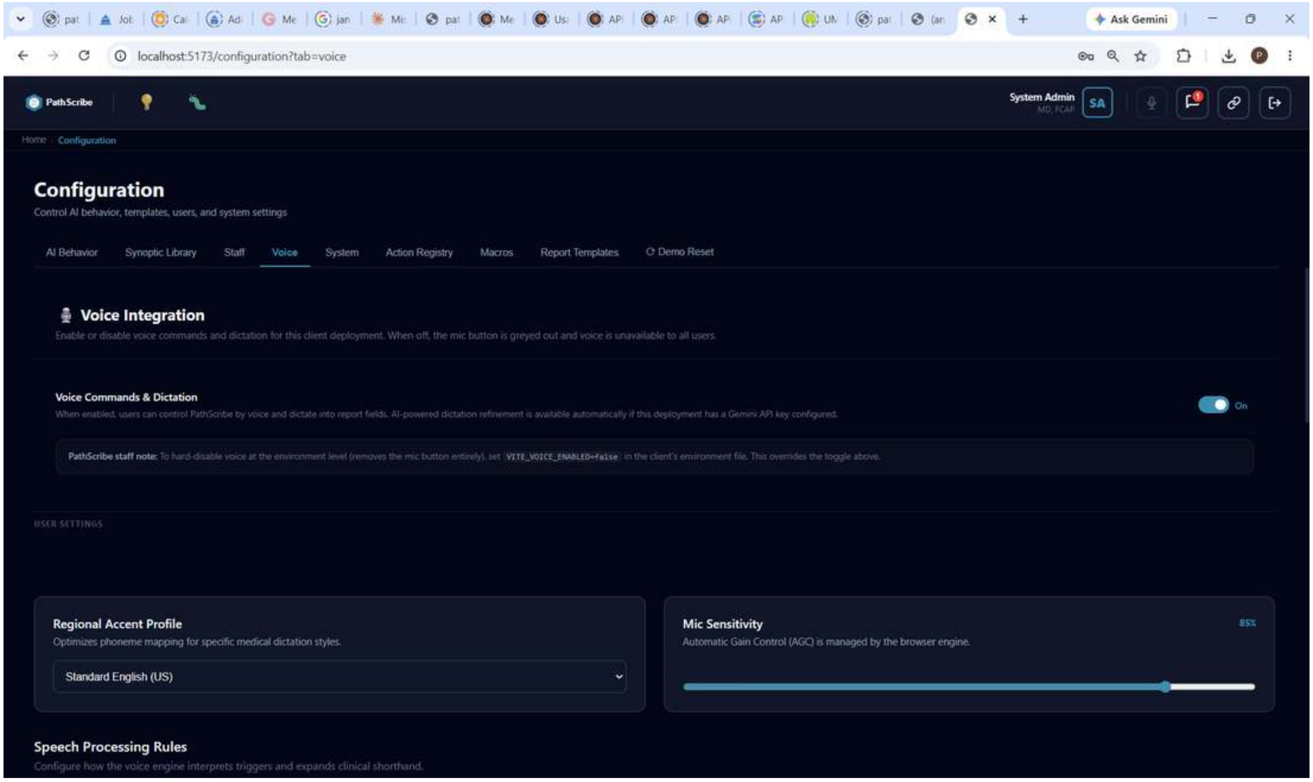
Build / Customise Synoptic Form.

17. Voice

Navigate to Configuration -> Voice. Distinct from Action Registry (Section 10), which maps voice phrases to in-app actions -- this tab controls voice/dictation availability and speech-to-text shorthand expansion.

17.1 Voice Integration

Master switch for voice commands and dictation across this client deployment. When off, the mic button is greyed out for all users.



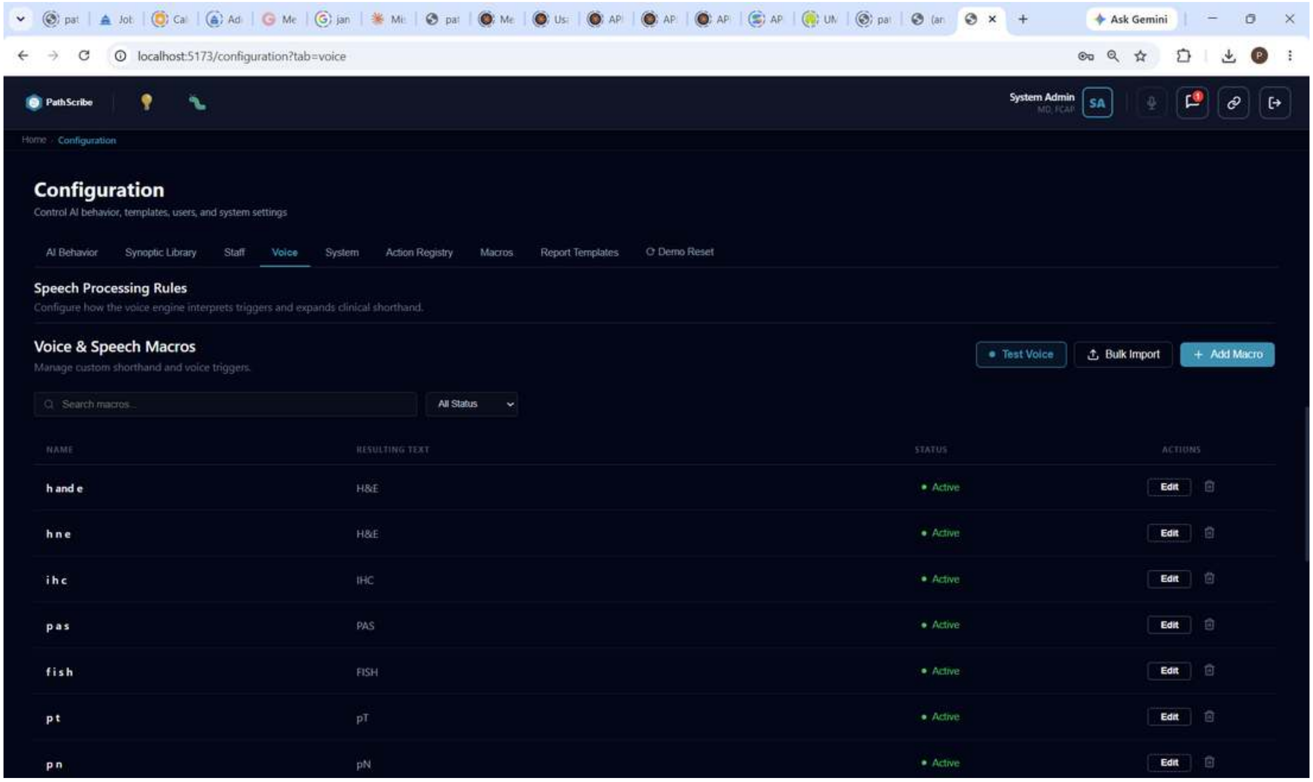
Voice Integration -- master toggle, Regional Accent Profile, Mic Sensitivity.

> **PathScribe staff note**
>
> To hard-disable voice at the environment level (removing the mic button entirely, overriding the toggle above), set VITE_VOICE_ENABLED=false in the client's environment file.

AI-powered dictation refinement is available automatically if this deployment has a Gemini API key configured. **Regional Accent Profile** optimises phoneme mapping for specific medical dictation styles. **Mic Sensitivity** tunes input gain (automatic gain control is managed by the browser engine).

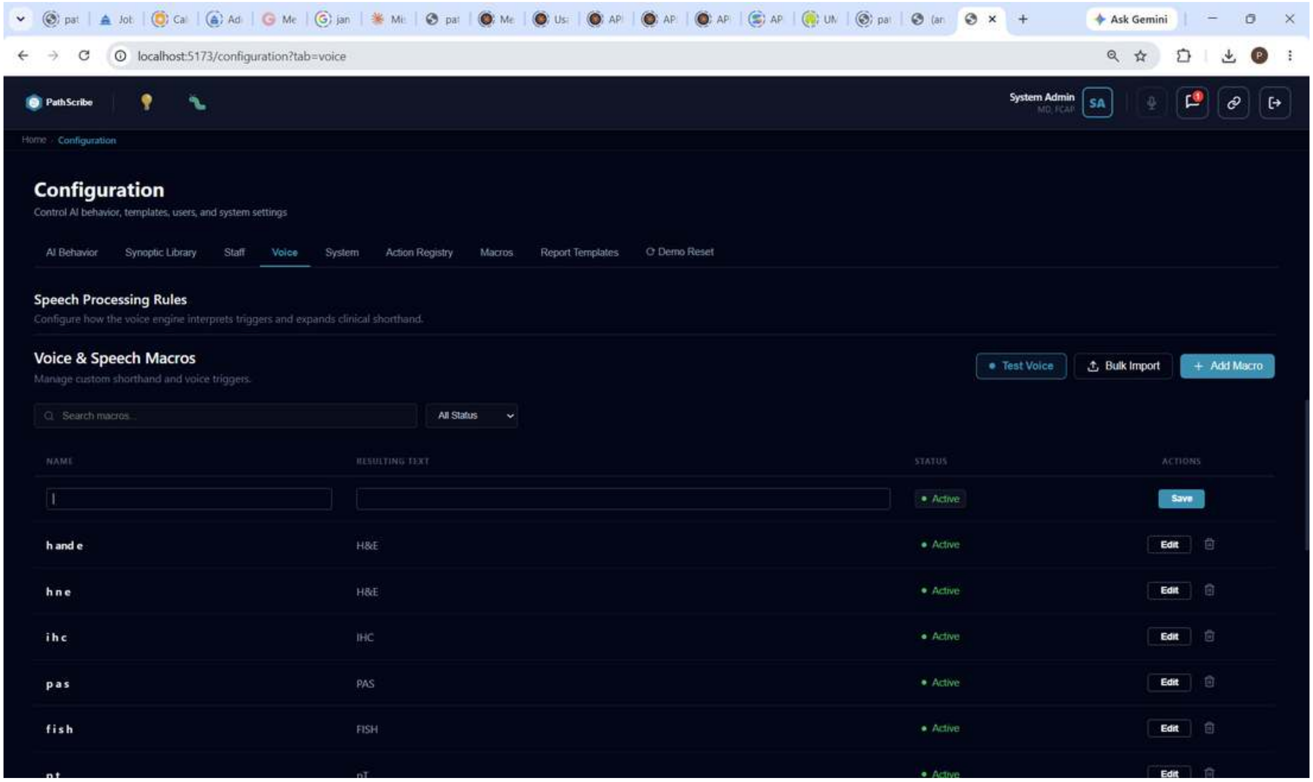
17.2 Voice & Speech Macros

Distinct from the text-snippet Macros tab (Section 15) -- these solve dictation mis-hearing common pathology abbreviations by mapping a spoken letter sequence to the correct resulting text (e.g. saying "h and e" or "h n e" dictates as H&E; "p t" dictates as pT).

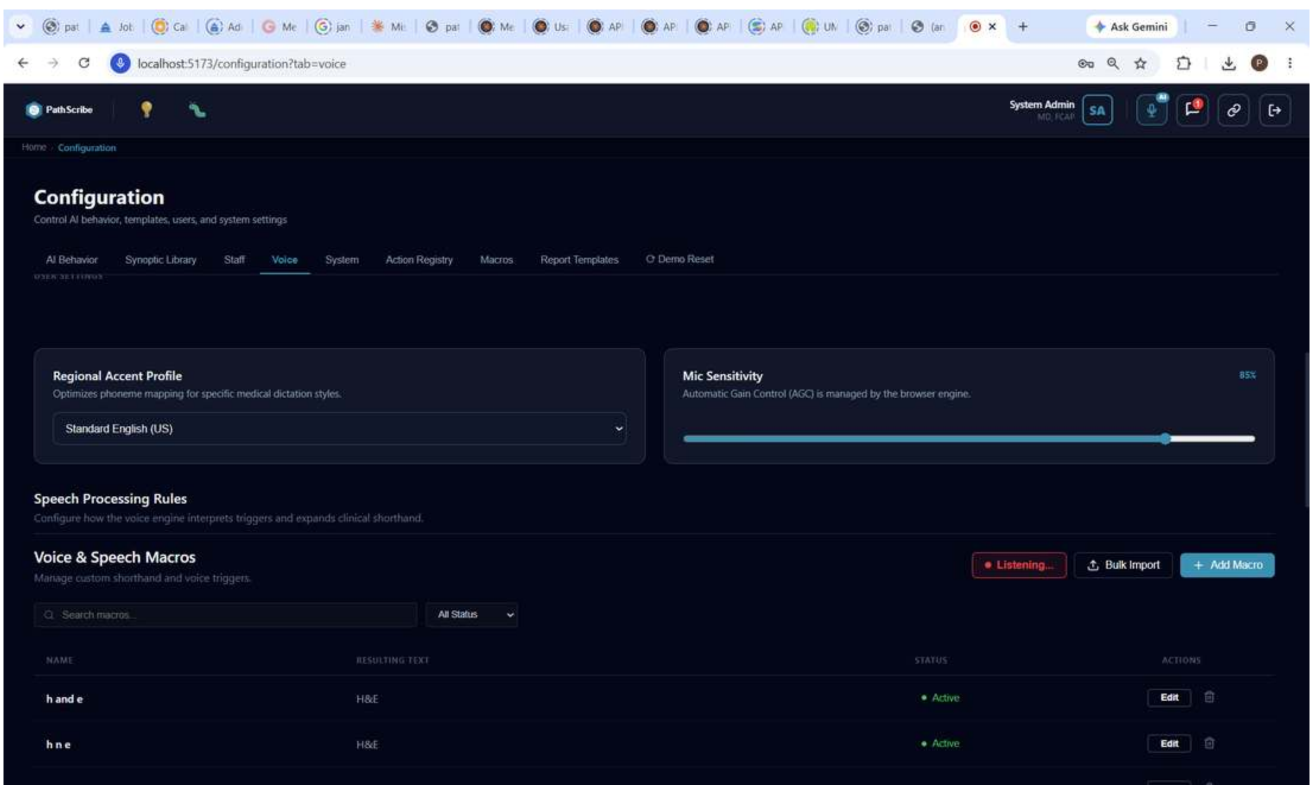


Voice & Speech Macros -- spoken trigger to resulting text, with Test Voice and Bulk Import.

Use **Test Voice** to verify recognition live, **Bulk Import** to load many macros at once, and **+ Add Macro** for a single new entry. Each macro can be toggled Active/Inactive without deleting it.



+ Add Macro -- an inline row with Name and Resulting Text fields, no separate modal.



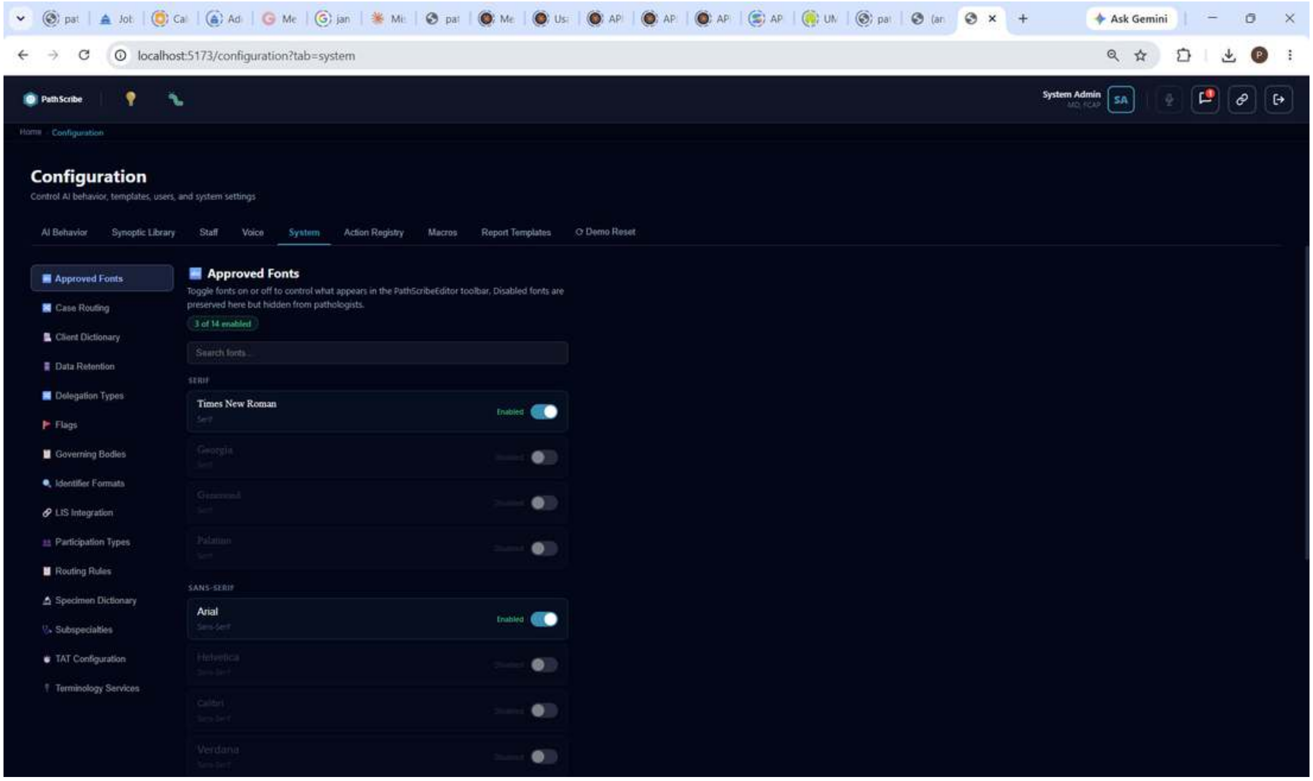
Test Voice -- the mic icon and Test Voice button both switch to a live "Listening..." state.

18. System Tab -- Additional Settings

The System Tab sidebar has 15 sections in total. Sections 7.1-7.6 (LIS Integration, Jurisdiction, Identifier Formats, Specimens, Subspecialties, Terminology Services) are covered in Section 7. The remaining sections are covered here. Case Routing is intentionally not yet documented in this guide -- pending a Standalone/Orchestration case creation workflow that doesn't exist yet; see your engineering team for current status.

18.1 Approved Fonts

Controls which fonts appear in the PathScribeEditor toolbar (the rich text editor used for Gross/Microscopic/Macro content). Toggle each font on or off; disabled fonts are preserved here but hidden from pathologists. 14 fonts are available across Serif and Sans-Serif families.

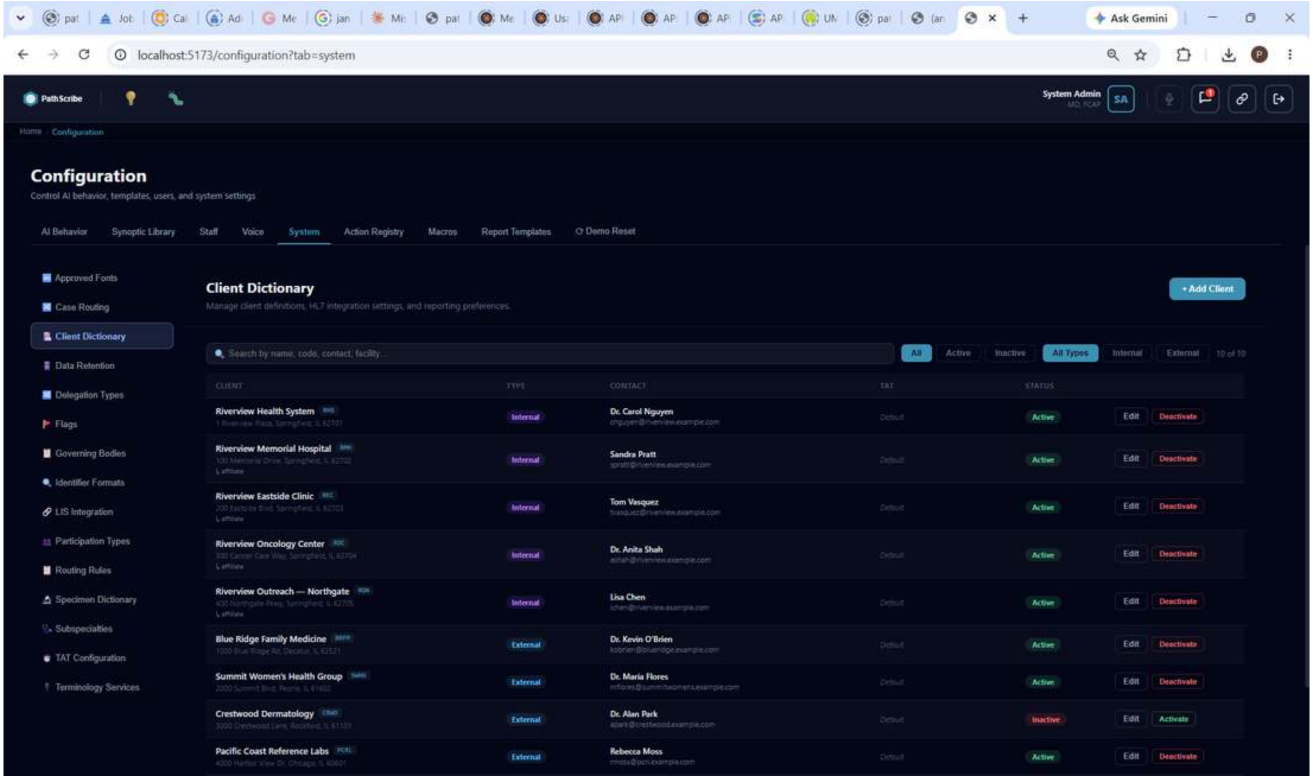


Approved Fonts -- per-font enable/disable toggle, grouped by family.

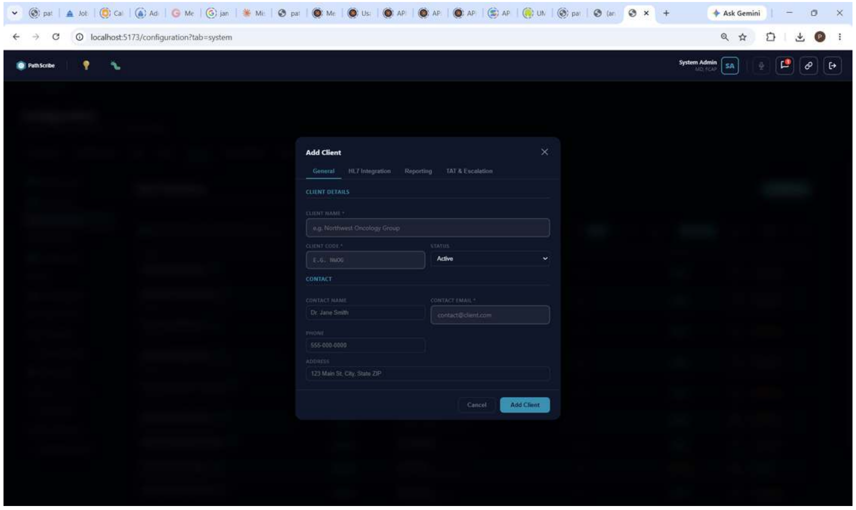
Use the search box to filter the list by font name when the family groups get long.

18.2 Client Dictionary

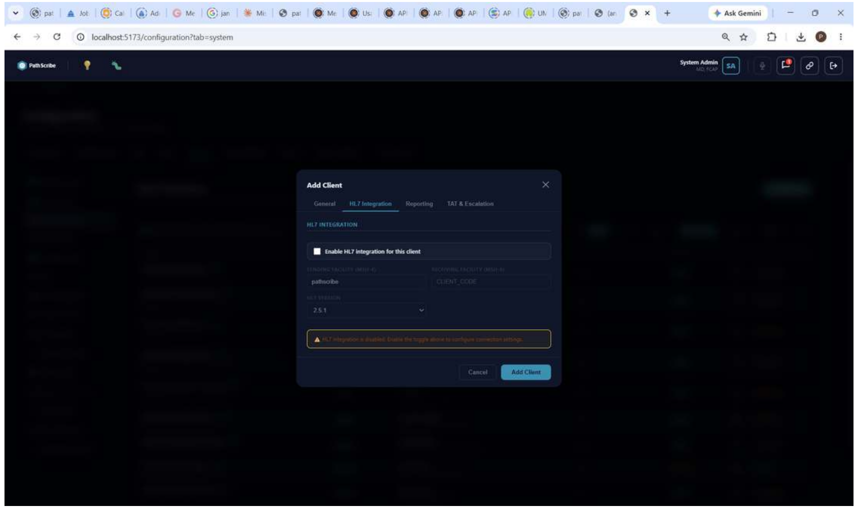
Manages client (submitting facility) definitions, HL7 integration settings, and reporting preferences. Clients are tagged Internal or External, with status (Active/Inactive) and an optional TAT override.


Client Dictionary -- Internal/External clients with contact and status.

Add Client is a four-tab modal: **General** (name, code, contact, address, status), **HL7 Integration**, **Reporting**, and **TAT & Escalation**.

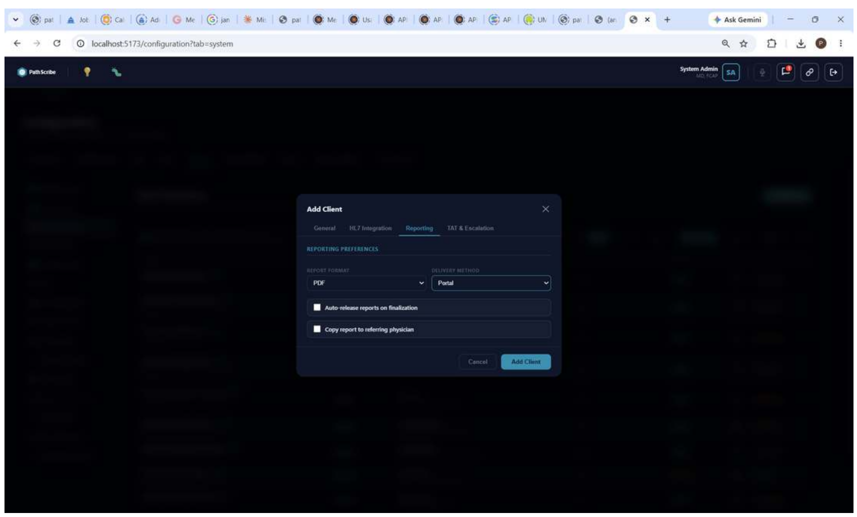

Add Client -- General tab.

HL7 Integration: per-client toggle, Sending Facility (MSH-4) and Receiving Facility (MSH-6) codes, and HL7 version (e.g. 2.5.1).



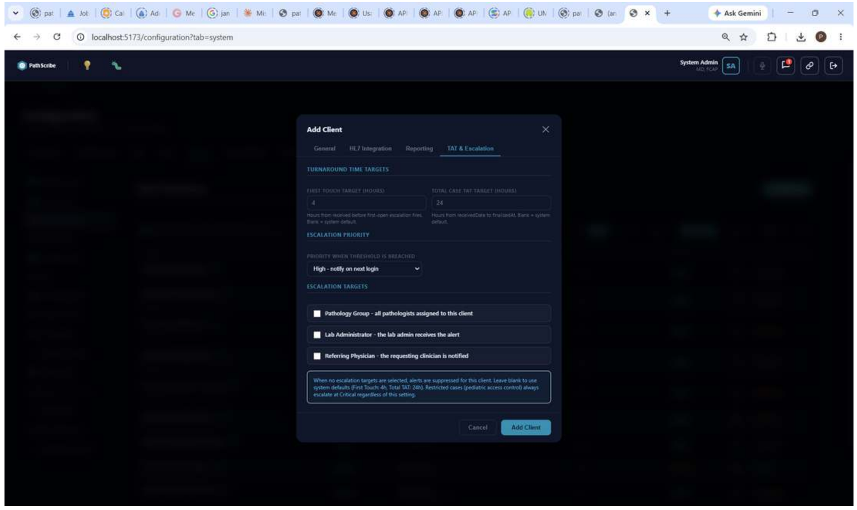
Add Client -- HL7 Integration tab.

Reporting: Report Format (e.g. PDF) and Delivery Method (e.g. Portal), plus Auto-release reports on finalization and Copy report to referring physician toggles.



Add Client -- Reporting tab.

TAT & Escalation: per-client First Touch and Total Case TAT targets (overriding the system default), an escalation priority, and which roles get notified on breach (Pathology Group, Lab Administrator, Referring Physician).

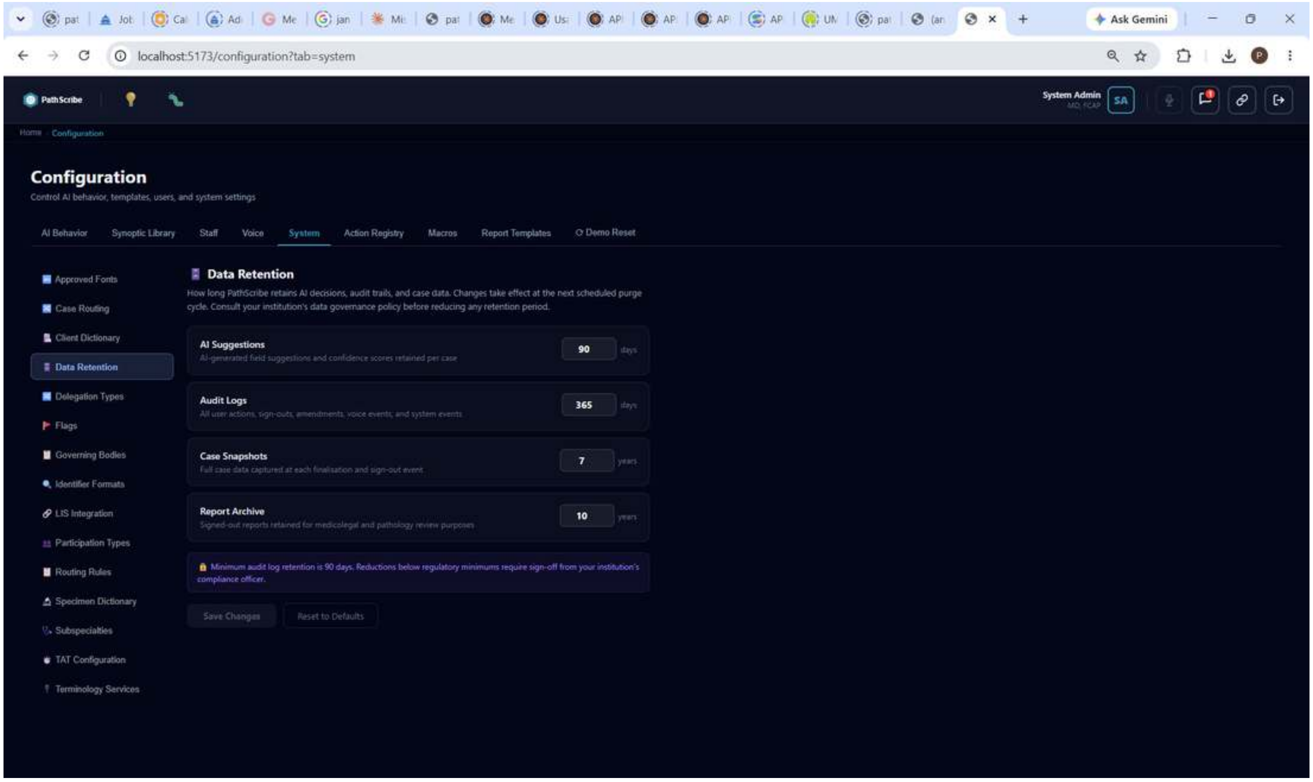


Add Client -- TAT & Escalation tab.

> **NOTE**
>
> Pediatric-restricted cases always escalate at Critical severity regardless of this client's escalation settings (Section 2.4, Paediatric Access Control).

18.3 Data Retention

Controls how long PathScribe retains AI decisions, audit trails, and case data. Changes take effect at the next scheduled purge cycle.



Data Retention -- AI Suggestions, Audit Logs, Case Snapshots, Report Archive.

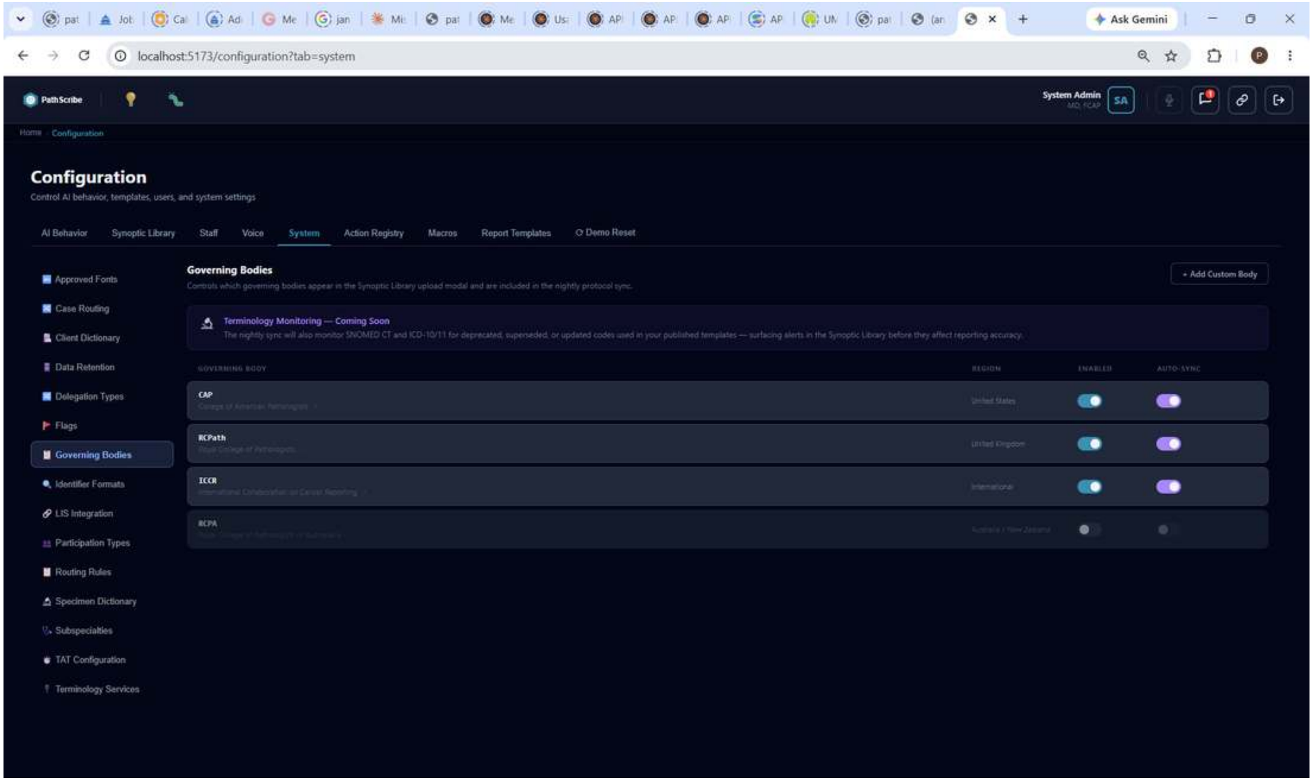
Setting	Default	Description
AI Suggestions	90 days	AI-generated field suggestions and confidence scores retained per case.
Audit Logs	365 days	All user actions, sign-outs, amendments, voice events, and system events.
Case Snapshots	7 years	Full case data captured at each finalisation and sign-out event.
Report Archive	10 years	Signed-out reports retained for medicolegal and pathology review purposes.

KNOWN LIMITATION (as of this writing)

Minimum audit log retention is 90 days. Reductions below regulatory minimums require sign-off from your institution's compliance officer.

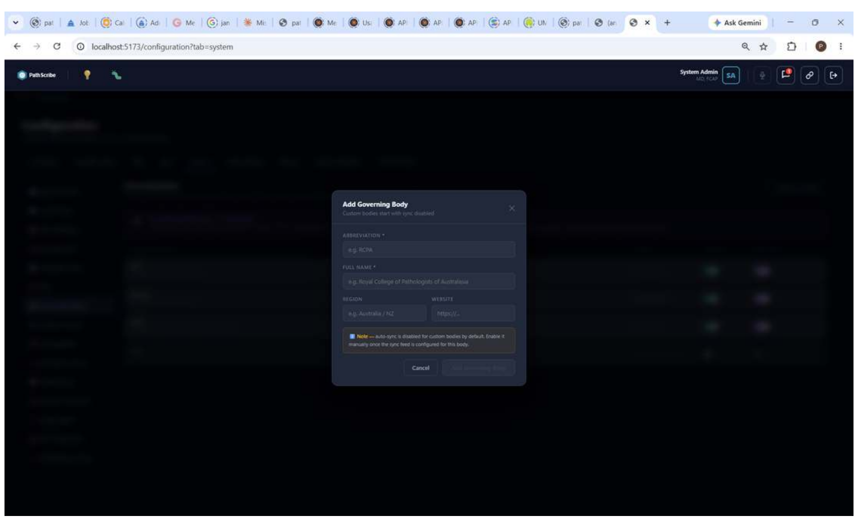
18.4 Governing Bodies

Controls which governing bodies appear in the Synoptic Library upload modal and are included in the nightly protocol sync. Each body has independent Enabled and Auto-Sync toggles.



Governing Bodies -- CAP, RCPath, ICCR active; RCPA disabled. Auto-sync per body.

Seeded bodies: CAP (United States), RCPath (United Kingdom), ICCR (International), and RCPA (Australia / New Zealand). Click + Add Custom Body to add another: Abbreviation, Full Name, Region, and Website.

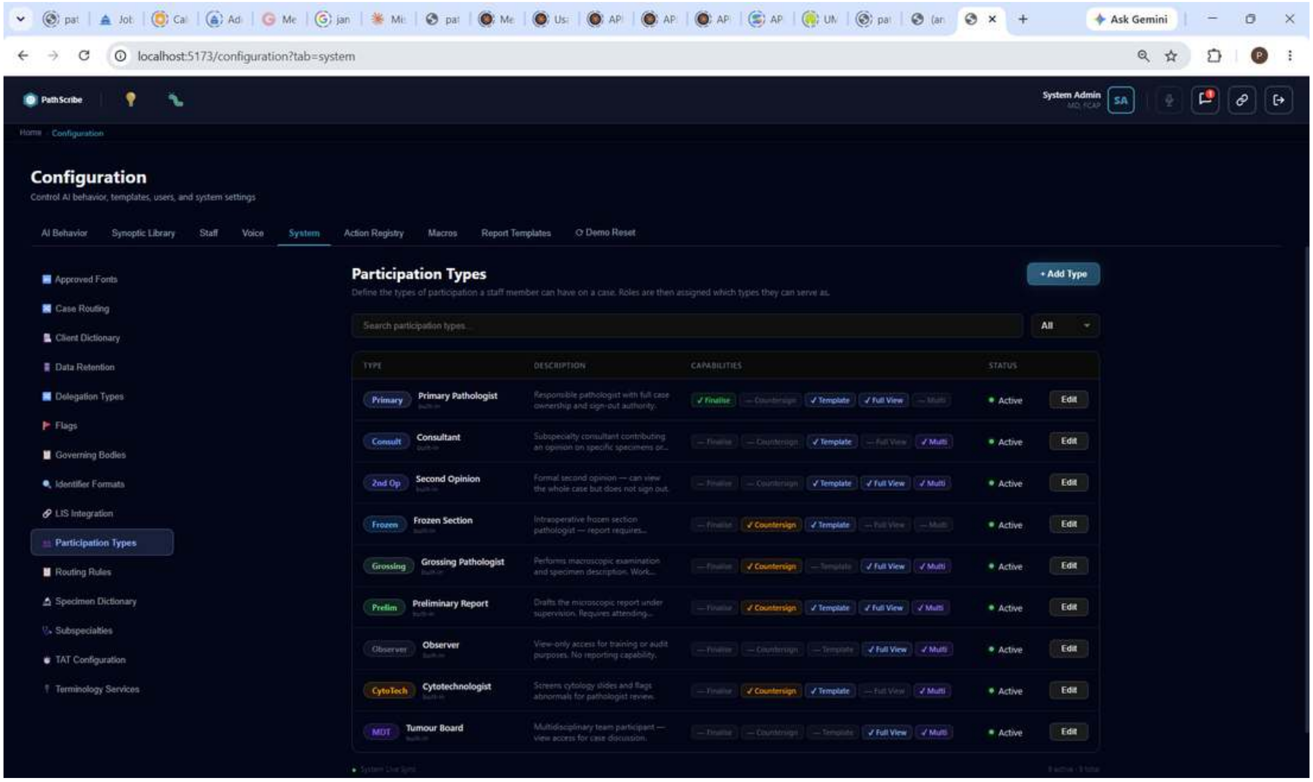


Add Governing Body -- custom bodies start with sync disabled.

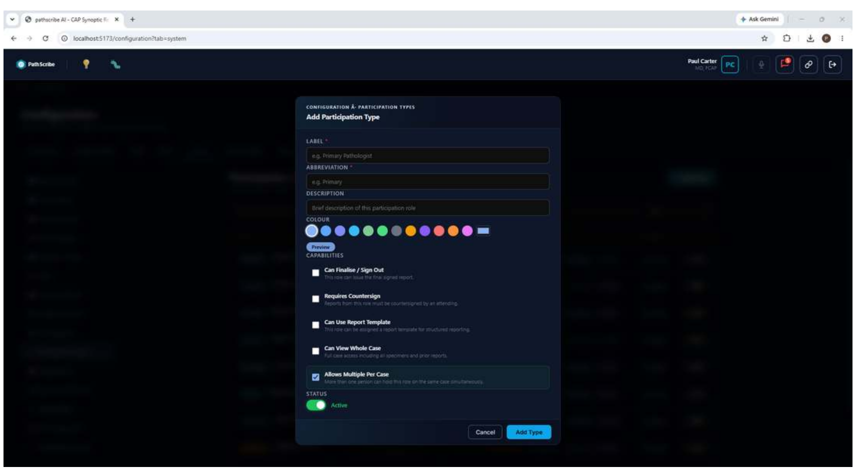
18.5 Participation Types

Defines the types of participation a staff member can have on a case; Roles are then assigned which types they can serve as. Each type carries its own capability flags -- Can Finalise/Sign Out, Requires Countersign, Can Use Report Template, Can View Whole Case, Allows Multiple Per Case -- so permissions are the intersection of

Role and Participation Type, never a single flat role.



Participation Types -- Primary, Consult, 2nd Op, Frozen, Grossing, Prelim, Observer, CytoTech, MDT, each with its own capability flags.



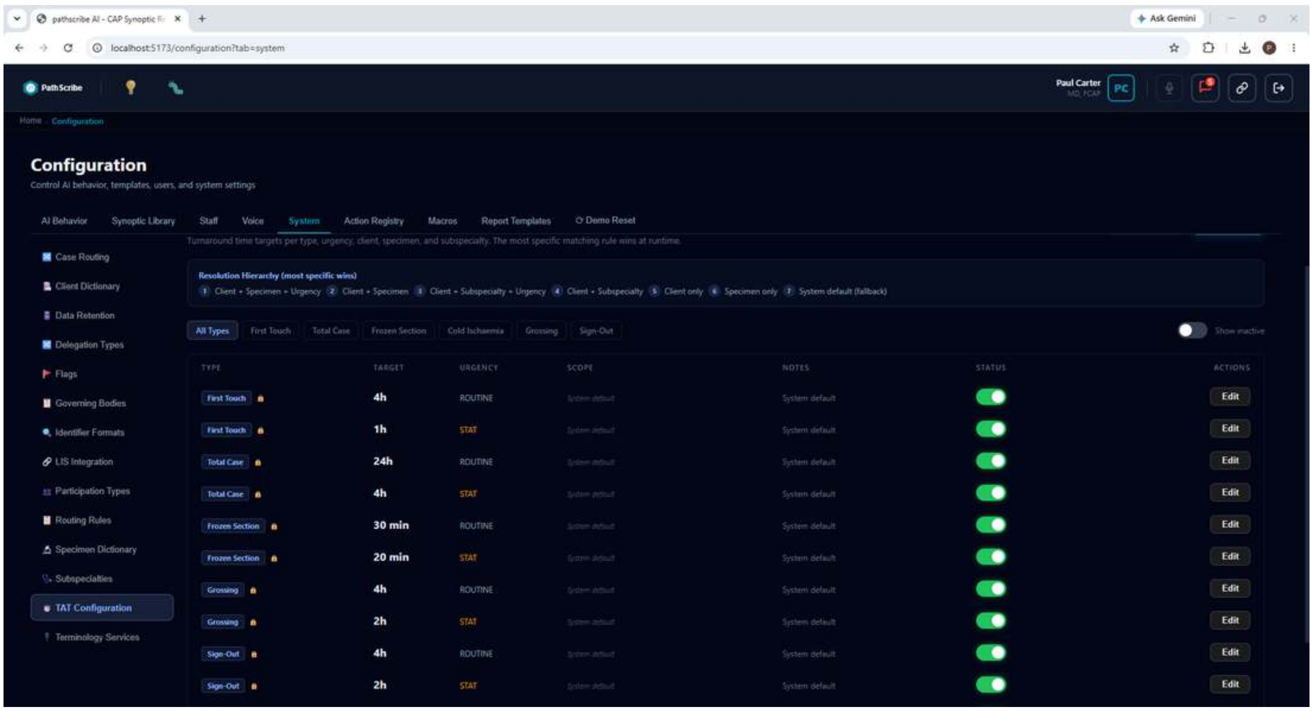
Add Participation Type -- Label, Abbreviation, Colour, and the five capability checkboxes.

18.6 TAT Configuration

Turnaround time targets per type, urgency, client, specimen, and subspecialty. The most specific matching rule wins at runtime, via a seven-level resolution hierarchy:

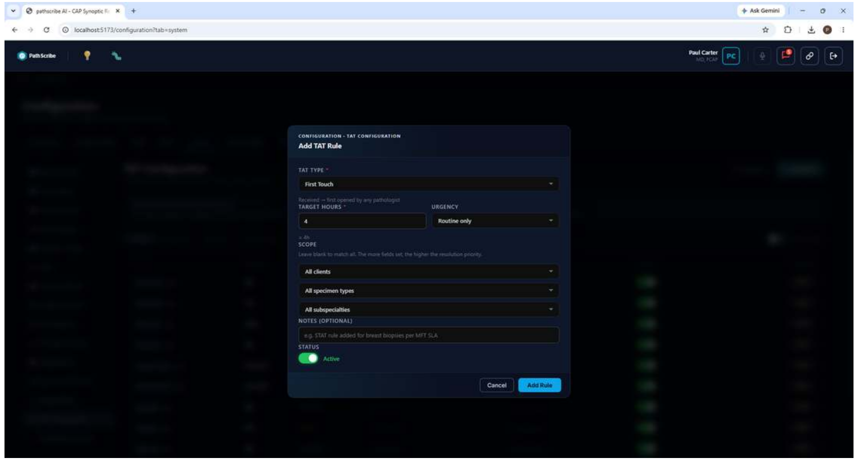
1. Client + Specimen + Urgency

2. Client + Specimen
3. Client + Subspecialty + Urgency
4. Client + Subspecialty
5. Client only
6. Specimen only
7. System default (fallback)



TAT Configuration -- resolution hierarchy and per-type targets (First Touch, Total Case, Frozen Section, Grossing, Sign-Out), each split by Routine/STAT urgency.

Click + Add Rule: TAT Type, Target Hours, Urgency, and Scope (Client / Specimen Type / Subspecialty -- leave any blank to match all). The more scope fields set, the higher the resolution priority, matching the hierarchy above.



Add TAT Rule.

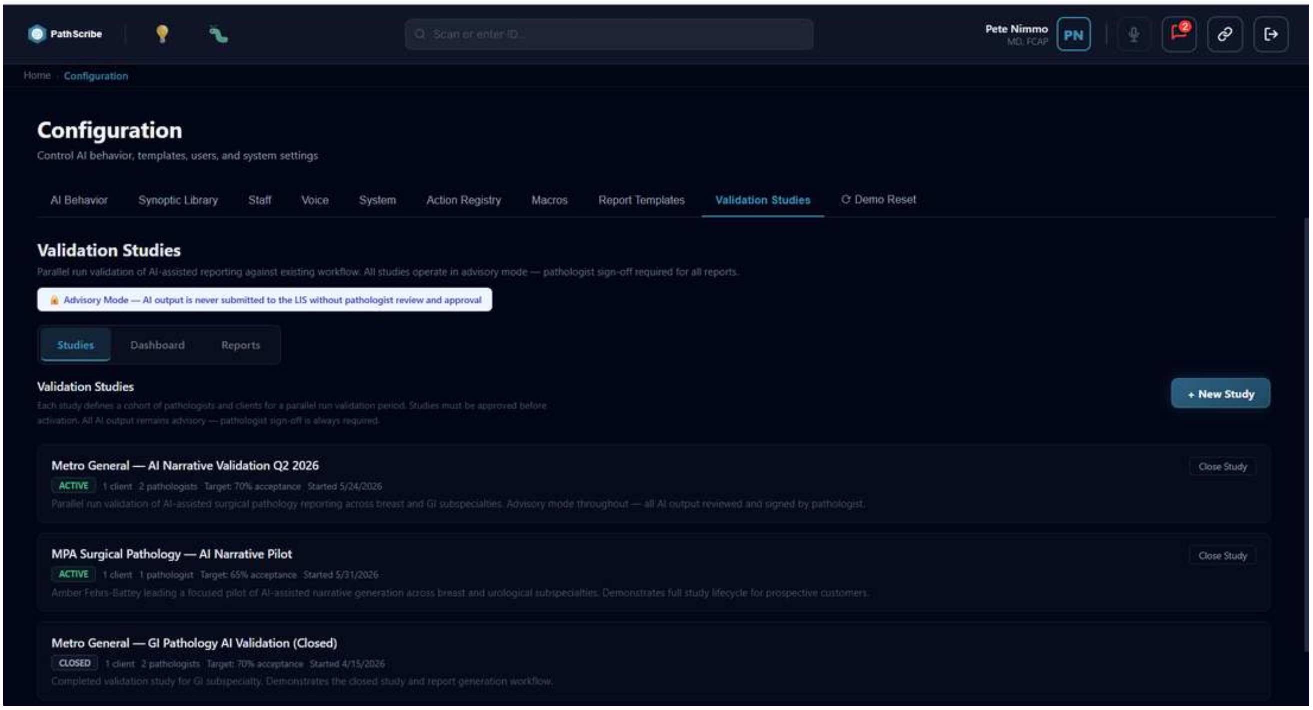
19. Validation Studies

Navigate to Configuration -> Validation Studies. Validation Studies run a parallel-run validation of AI-assisted reporting against your existing workflow, measuring how often AI-drafted narrative sections are accepted unchanged versus edited. Three sub-tabs: Studies, Dashboard, Reports.

> **NOTE**
>
> Advisory Mode -- AI output is never submitted to the LIS without pathologist review and approval. This applies for the full duration of any validation study, with no exception.

19.1 Studies

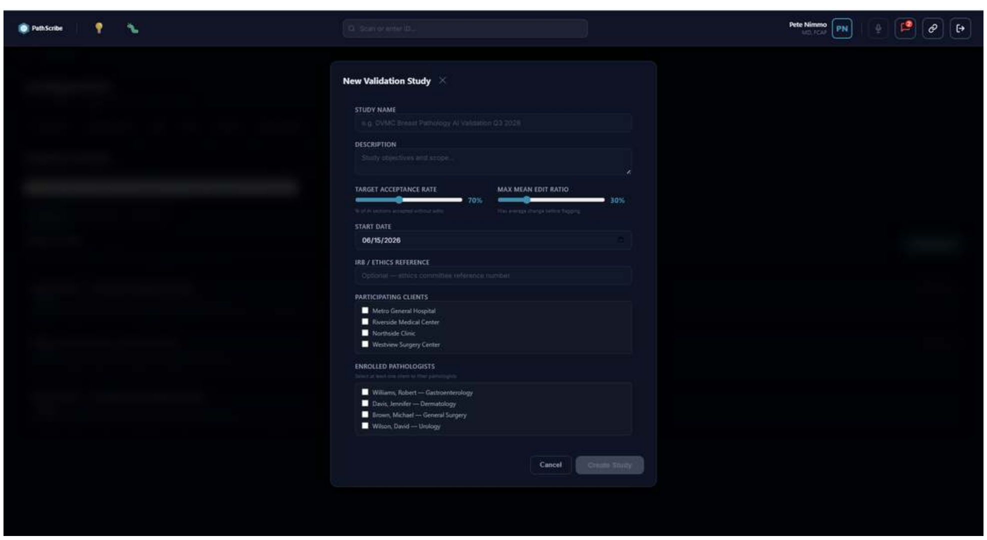
Each study defines a cohort of pathologists and clients for a parallel-run validation period, with a target acceptance rate and a start date. Studies must be approved before activation. Status badges: ACTIVE or CLOSED. Click Close Study to end an active study and move it into Reports.



Validation Studies sits in the same Configuration tab bar as AI Behavior, Synoptic Library, Macros, and Report Templates -- an admin-configured feature, not a pathologist workflow screen.

19.2 Creating a Study

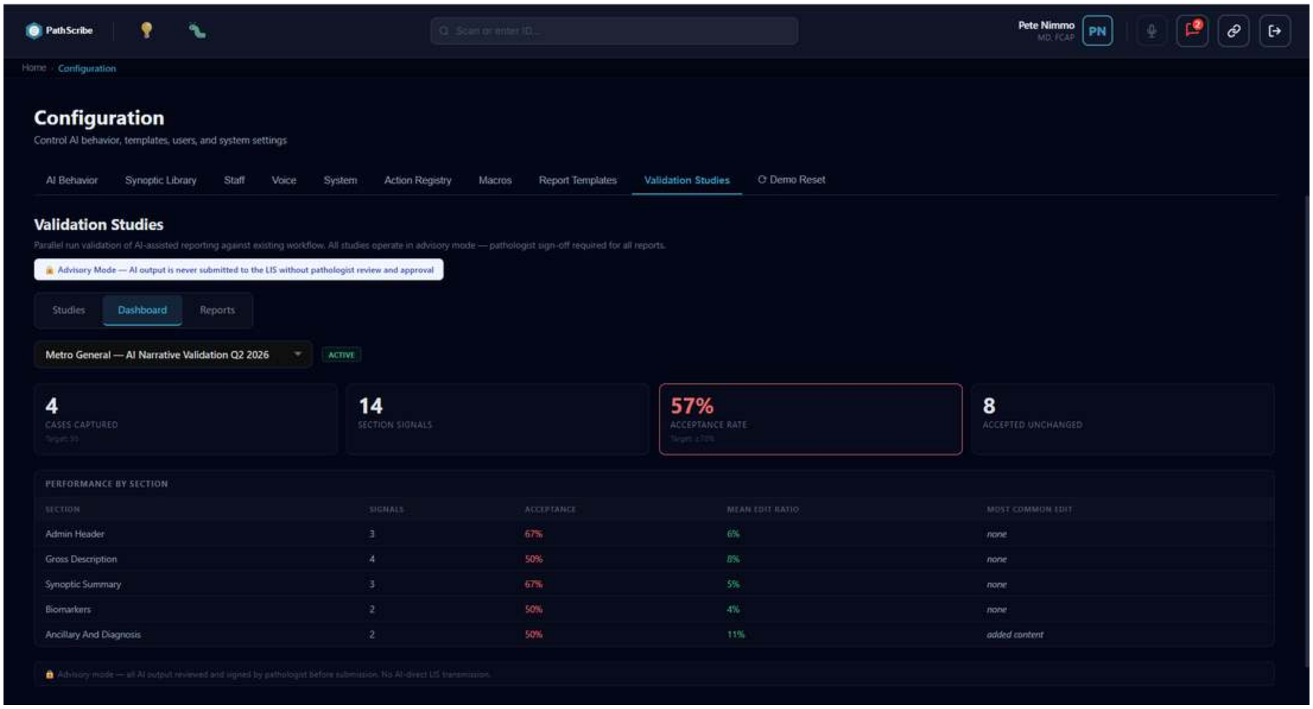
Click + New Study. Key fields: Study Name, Description, Target Acceptance Rate and Max Mean Edit Ratio (both configurable thresholds), Start Date, an optional IRB / Ethics Reference, Participating Clients, and Enrolled Pathologists (filtered by the clients selected).



New Validation Study -- thresholds, participating clients, enrolled pathologists.

19.3 Dashboard

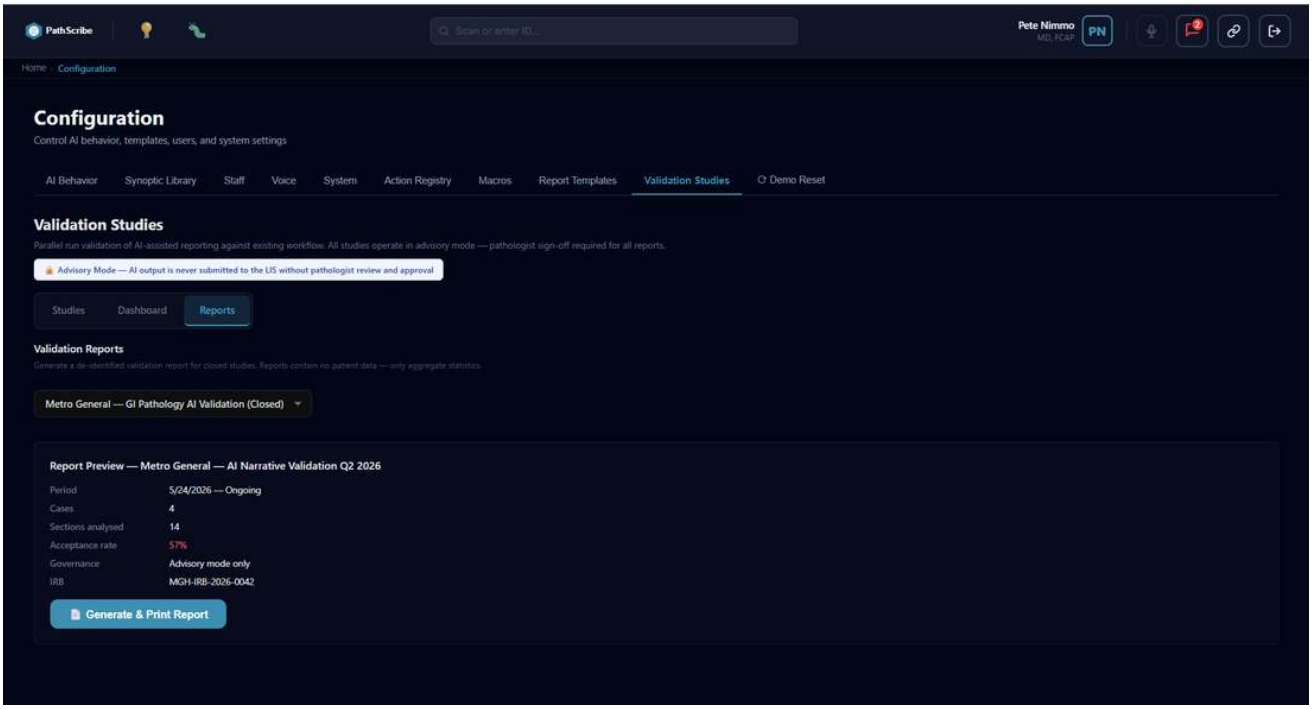
Live metrics for the active study: Cases Captured (against a target), Section Signals, Acceptance Rate (against its target threshold), and Accepted Unchanged. A Performance-by-Section breakdown shows signals, acceptance rate, mean edit ratio, and the most common edit type per narrative section -- useful for spotting which sections need AI instruction tuning before relying on them more heavily.



Validation Studies Dashboard -- live acceptance metrics by section.

19.4 Reports

Exportable study reports for compliance and quality review, including closed studies.



Validation Studies -- Reports tab.

Quick Reference -- Administration

Task	Navigation path
Add a staff user	Configuration -> Staff -> Add User
Deactivate a user	Configuration -> Staff -> [user] -> Deactivate
Manage physician directory	Configuration -> System -> Physicians
Edit role permissions	Configuration -> Staff -> Roles -> [role] -> Permissions
Create a flag	Configuration -> System -> Flags -> Add Flag
Configure computational flag	Configuration -> System -> Flags -> [flag] -> Computational
Add a delegation type	Configuration -> System -> Delegation Types -> + New Type
Open Template Builder	Configuration -> Template Builder
Create a new Part	Template Builder -> Part Library -> New Part
Assemble a template	Template Builder -> Templates -> New Template
Publish a template	Template Assembly page -> Publish
Change AI model	Configuration -> AI Settings -> Model
Set confidence threshold	Configuration -> AI Settings -> Confidence Threshold
Configure identifier formats	Configuration -> System -> Identifier Formats
Check terminology services	Configuration -> System -> Terminology Services
View action registry	Configuration -> Action Registry

Task	Navigation path
View audit log	Home -> System Audit
System information	Nav bar initials badge -> System Information
Generate support report	System Information -> Copy to Clipboard
Reset demo data	Configuration -> Demo Reset (development only)

Part II -- Report Template Configuration

Building, assembling, routing, and testing Report Templates.

STATUS

Text content in this Part has been verified directly against application code. Screenshots are pending -- bracketed placeholders below mark where each will be inserted.

1. Introduction

1.1 Purpose of This Guide

This guide explains, in full operational detail, how Report Templates are built, assembled, tested, and routed inside PathScribe's Orchestration mode. It is written for the customer administrator who is responsible for configuring report structure -- not for end-user pathologists, who interact with the finished output rather than its configuration.

Every setting, field, button, and behavior described here has been verified directly against the underlying application code, not inferred from the interface alone. Where a setting has a limitation, an edge case, or does not yet do what its label implies, that is stated explicitly rather than glossed over.

1.2 Who Should Read This Guide

- Create or modify the building blocks ("Parts") that make up a report
- Assemble Parts into complete Report Templates
- Configure which template is automatically selected for a given case, client, or physician
- Test and validate template routing before relying on it for real cases

1.3 Two Reporting Modes: CoPilot and Orchestration

- **CoPilot mode** -- the host Laboratory Information System (LIS) owns the report. PathScribe feeds structured synoptic data back to the LIS, which generates the final report using its own template. Report Templates as described in this guide are not used in CoPilot mode.
- **Orchestration mode** -- PathScribe owns the full report lifecycle for the case. The Report Template you configure here determines the structure of the document, and the Orchestrator Engine uses it (together with synoptic data) to AI-draft each section. This is the mode this entire guide is about.

> **NOTE**
>
> Throughout this guide, "report template" always refers to an Orchestration-mode document template (built from Parts, in the Report Templates tab) -- never to a CAP/RCPath synoptic checklist. Section 2.4 explains this distinction in detail.

2. Core Concepts

(Sections 2.1-2.3, 2.5-2.8 -- Parts, Templates, Routing Rules, the resolution cascade, and status lifecycle -- are covered in operational detail across Sections 4-6 below; this section establishes vocabulary referenced throughout.)

2.4 Synoptic Template vs. Report Template

- A **Synoptic Template** (CAP, RCPath, or Custom) is a structured data-entry checklist for one specimen type, configured in the Synoptic Library. It governs what discrete fields a pathologist fills in.
- A **Report Template** (this guide) is the document shell -- built from Parts -- that determines the structure, formatting, and narrative sections of the generated report itself.

Routing resolves which Report Template applies based partly on which Synoptic Template was used, among other signals (Section 6.1) -- but they remain two distinct systems, configured in two separate parts of the application.

3. Getting Started

Navigate to Configuration -> Report Templates (admin role required). The screen has two tabs: Part Library and Templates.

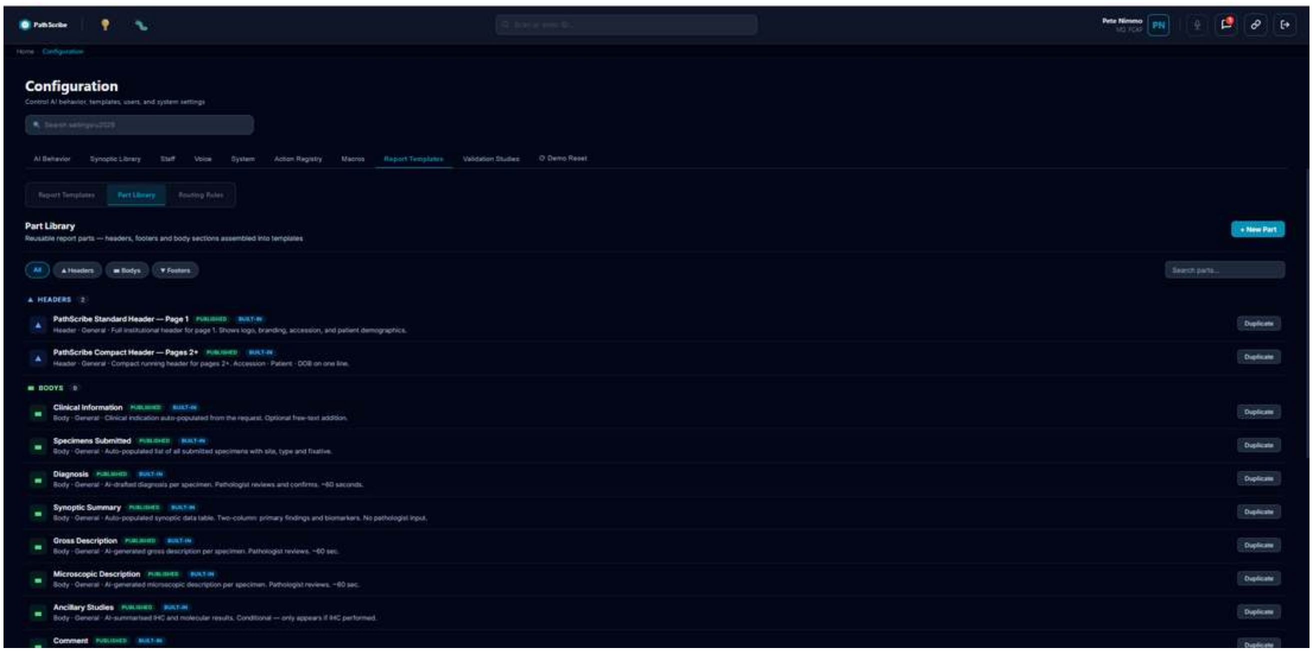
4. The Part Builder

4.1 Built-in (Protected) Parts

PathScribe ships with built-in Parts (see Appendix A for the complete list). None are editable directly -- duplicate first to customize (Section 4.10).

4.2 Browsing the Part Library

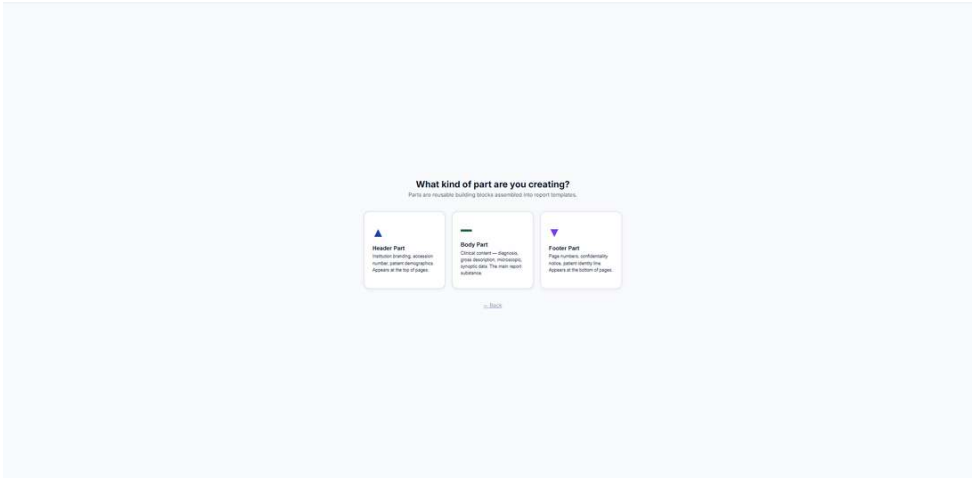
The Part Library list is grouped by type (Headers, Bodys, Footers) and supports filtering by type and free-text search. Each row shows name, specialty, description, status badge, and a BUILT-IN badge where applicable.



Part Library list view -- type filter pills and grouped Headers/Bodys.

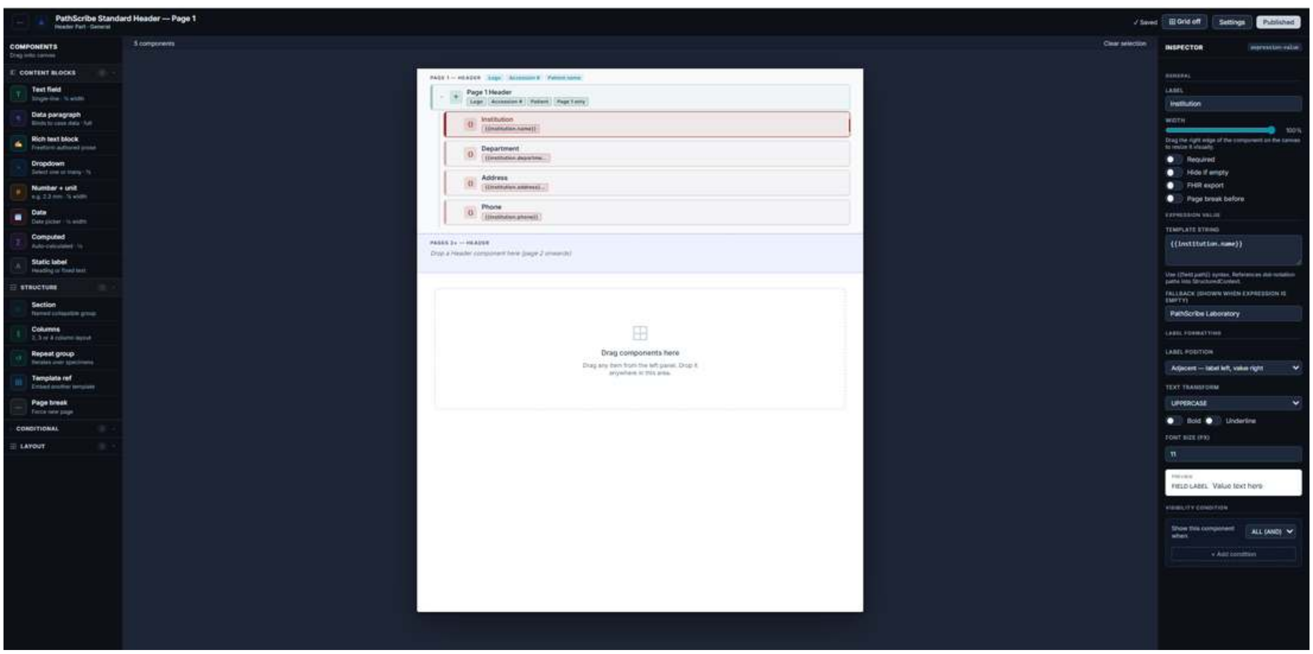
4.3 Build a New Reusable Part

- Click New Part in the Part Library tab.
- Enter a name, description, part type (Header / Body / Footer), specialty, and optional subspecialty.
- Set the Standard (CAP, RCPath, or Custom). Click Create to open the Part Builder canvas.



New Part type picker -- Header / Body / Footer.

4.5 Component Palette -- Every Type



Part Builder -- Components palette (left), Canvas (centre), Inspector (right).

Component	Description
Section	Collapsible group of related fields. Use for logical clinical groupings.
Text Field	Single-line field bound to a case data key. Renders as "Label: Value".
Data Paragraph	Multi-line text slot filled from case data. Rich text supported.
Rich Text Block	Static text identical in every report -- disclaimers, institutional statements.
Dropdown	Selection field with predefined options. Single or multi-select.
Number + Unit	Numeric input with unit label. Configurable min/max and decimal places.

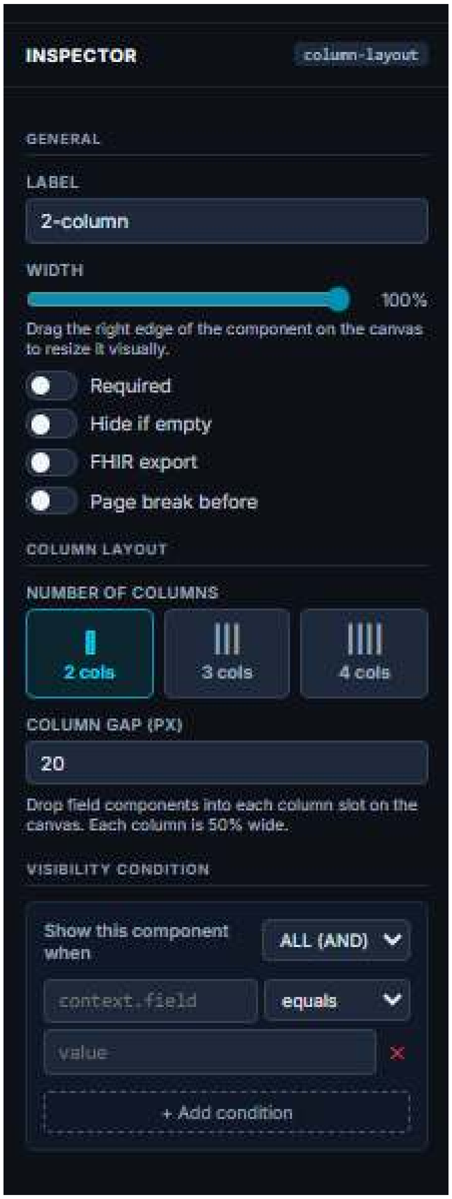
Component	Description
Date	Date field. Configurable format. Can default to today.
Computed	Calculated field derived from an expression combining other fields.
Static Label	Heading or label at any typographic level (H1-H3, body, caption).
Repeat Group	Container that iterates over a collection (e.g. specimens).
Column Layout	Arranges children in 2, 3, or 4 flowing columns.
If / Show When	Conditional container -- children shown only when a condition is true.
Page Break	Forces a page break in the printed report.
Header	Institutional letterhead. Configure elements and scope (page 1 / all pages).
Footer	Page footer with page numbers and confidentiality notice.

4.6 The Inspector Panel

The Inspector shows all configurable properties for the selected component: General (Label, Width, Required, Hide if Empty), component-specific settings (binding key, options, validation), Label Formatting, and Visibility Condition.

Inspector -- Label Formatting with live preview.

Inspector -- Visibility Condition expression builder.



Inspector -- Column Layout component settings.

4.7 Part Settings Panel

Type (Header / Body / Footer), Specialty, Description, Standard (CAP / RCPath / Custom).

4.9 Save / Publish a Part

An auto-save indicator in the topbar shows "Unsaved," "Saving...," or "Saved." Click Publish when ready -- this is one-directional.

4.10 Duplicating a Part to Customize It

Built-in and published Parts cannot be edited directly. Duplicate first, then edit the duplicate.

4.11 Known Limitations

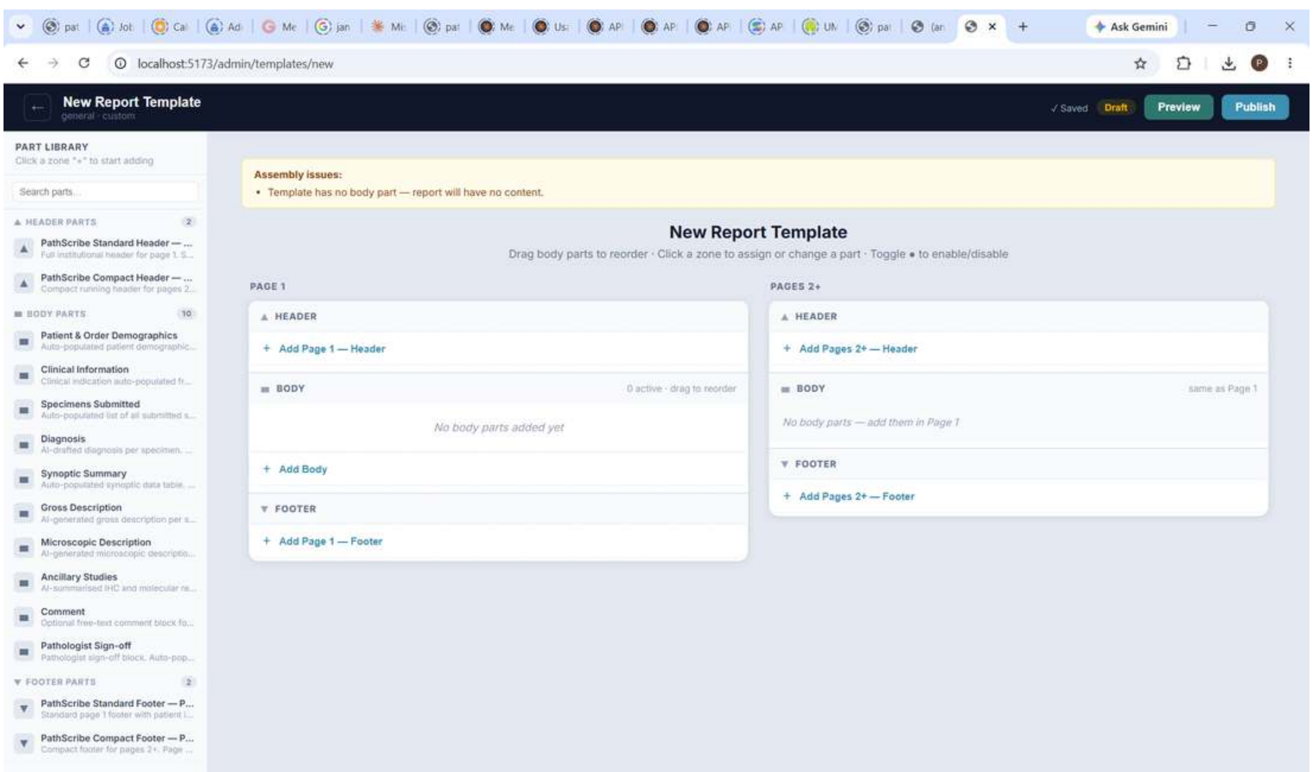
> **KNOWN LIMITATION (as of this writing)**
>
> There is currently no in-app way to revert a published Part back to Draft. If a published Part needs significant rework, the safer path is to duplicate it, edit the duplicate, and re-point templates at the duplicate once it's ready, rather than editing the published version live.

> **KNOWN LIMITATION (as of this writing)**
>
> If a Part's name is changed after it has already been added to one or more Report Templates, those templates' Assembly views continue showing the old name until the slot is removed and re-added. The underlying content is not affected -- only the displayed name in the Template Assembly screen goes stale.

5. Assembling Templates

5.1 Creating a New Template

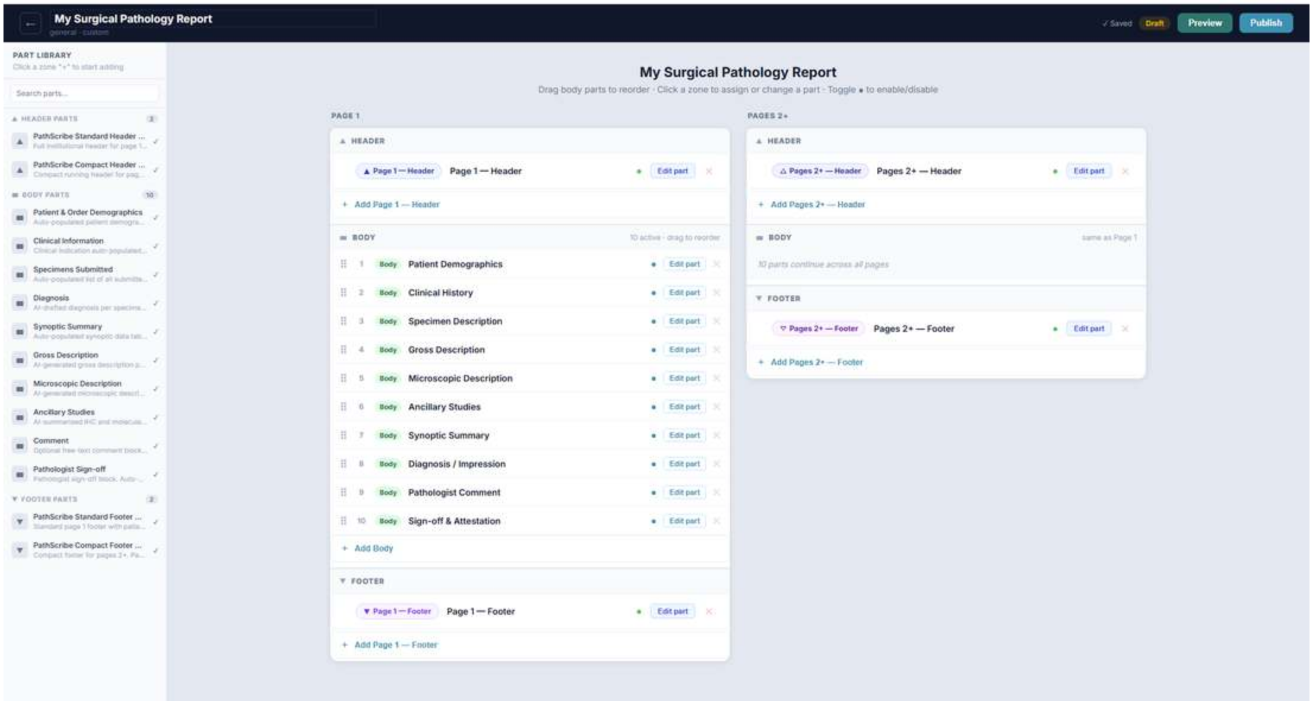
Two entry points, both in the Report Templates list view: **+ New Template** starts completely blank; **From Standard** clones the Gold Standard template's full slot configuration as a starting point, which is the faster path for most new subspecialty templates.



New Template, started blank -- no Body parts yet.

REAL-TIME VALIDATION

A blank template shows a live "Assembly issues: Template has no body part -- report will have no content" warning immediately on the canvas, before any attempt to publish -- not only at publish time.



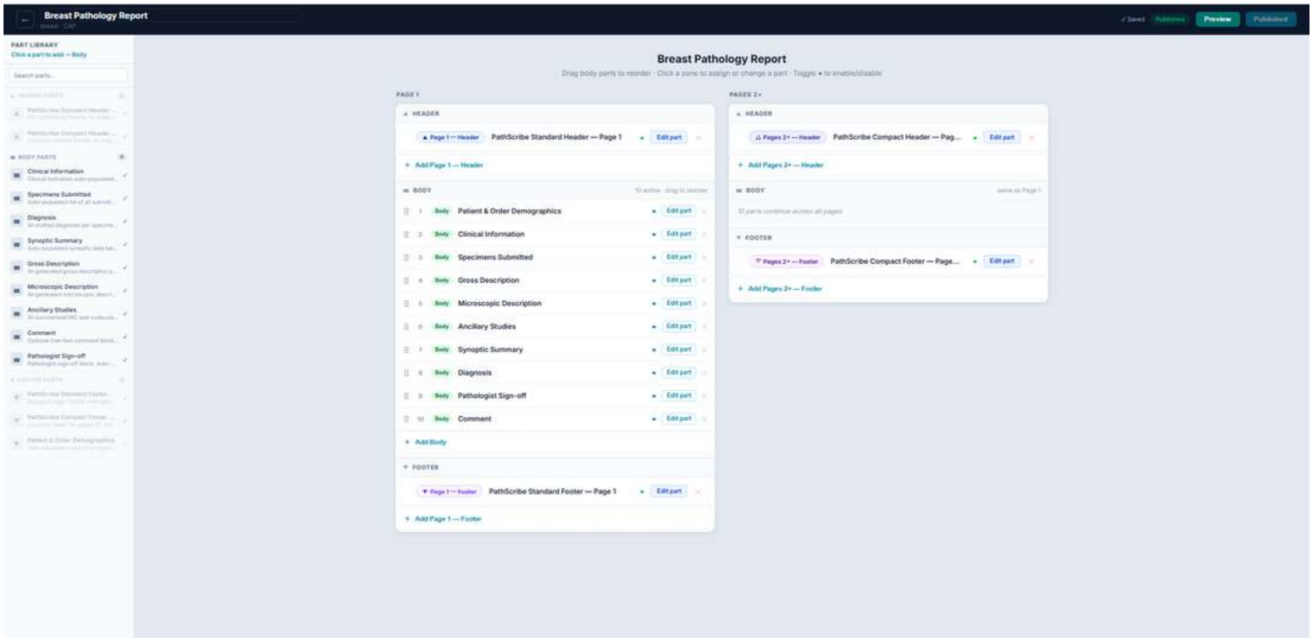
New Template via From Standard -- all 10 Body Parts already in place.

5.3 Naming a Template

Click the template name in the topbar (pencil/hover to edit).

5.4 Adding Parts to a Template

The Template Assembly page has two panels: left -- Part Library (drag parts into the assembly area); right -- Assembly Canvas with role zones: Page 1 Header, Pages 2+ Header, Body, Page 1 Footer, Pages 2+ Footer.



Adding a Body Part from the Part Library sidebar.

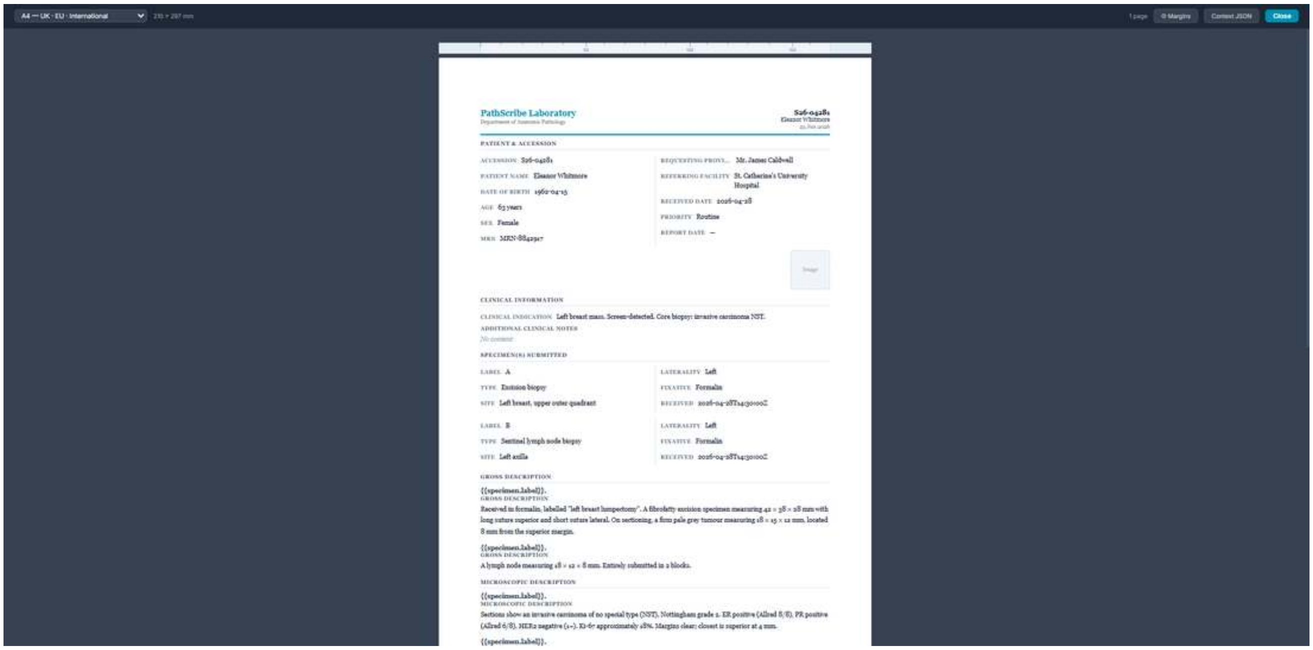
> **IMPORTANT**
>
> Minimum valid template: at least one Body Part is required to publish.

5.5 Reordering, Enabling, and Removing Slots

- Drag handle -- reorders Body slots only.
- Enable/disable dot -- skips a slot during generation without removing it.
- Edit part -- opens the Part Builder for that slot's Part.
- Remove -- removes the slot; the Part itself is unaffected.

5.7 Previewing a Template

Click Preview to render the template against mock data. Use the page size and margin controls to check across different print formats. A Context JSON panel shows exactly what mock data is available to each binding.



Preview panel -- page size selector, margins, Context JSON, and the fully rendered mock report.

5.8 Publishing a Template

Click Publish when the assembly is complete. The template becomes immediately available for selection in the synoptic reporting workflow and to Routing Rules. In-progress cases retain the version that was active when they were started. Publishing is one-directional.

5.9 Known Limitations

> **KNOWN LIMITATION (as of this writing)**
>
> As with Parts, there is no in-app "unpublish" action for a Report Template. Plan accordingly before publishing a template you expect to need to pull back from active use.

> **KNOWN LIMITATION (as of this writing)**
>
> If a Part used in a template is renamed after being added, the Assembly view's slot rows continue to show the name as it was at the moment the Part was added, not its current name. Removing and re-adding the slot refreshes it. This does not affect what is actually rendered in Preview or in a real generated report -- only the name shown in this configuration screen.

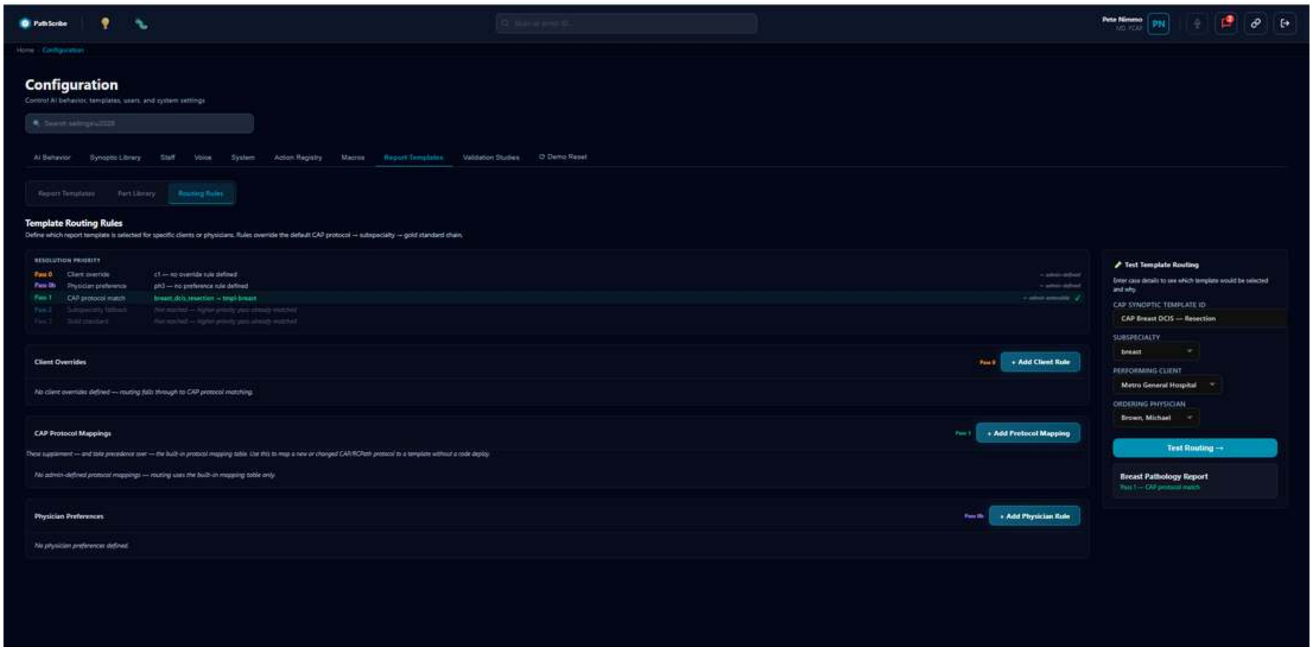
6. Routing Rules

6.1 The Five-Pass Resolution Cascade

Routing is a priority cascade, not an independent checklist -- once a pass matches, lower-priority passes are never evaluated:

- **Pass 0 -- Client Override:** a specific institution always gets a designated template.
- **Pass 0b -- Physician Preference:** a specific ordering physician always gets a designated template.

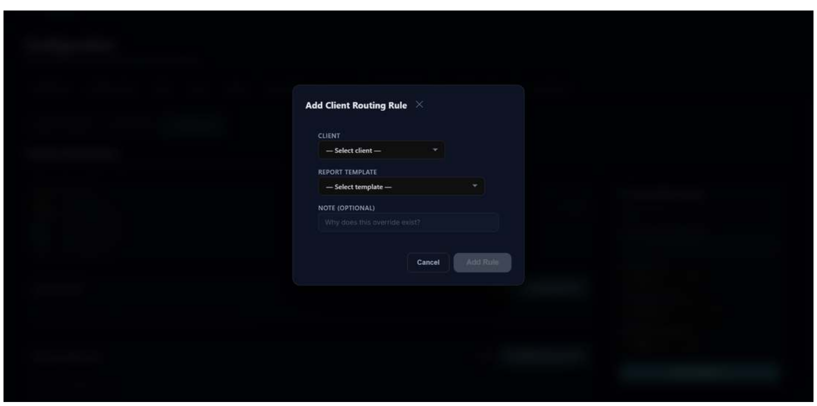
- **Pass 1 -- CAP Protocol Match:** the CAP/RCPath synoptic protocol ID maps to a designated template. Admin-defined mappings (Section 6.3) take precedence over the built-in mapping table when both exist for the same protocol.
- **Pass 2 -- Subspecialty Fallback:** a subspecialty value maps to a designated template.
- **Pass 3 -- Gold Standard:** the universal fallback if nothing else matched.



Template Routing Rules -- resolution priority list, Client Overrides, CAP Protocol Mappings, Physician Preferences, and the Test Template Routing panel, all in one screen.

6.2 Client Overrides

Navigate to Routing Rules -> Client Overrides -> + Add Client Rule. Select the client and the target (already-Published) template. Save.



Add Client Routing Rule -- select client, target Report Template, optional note.

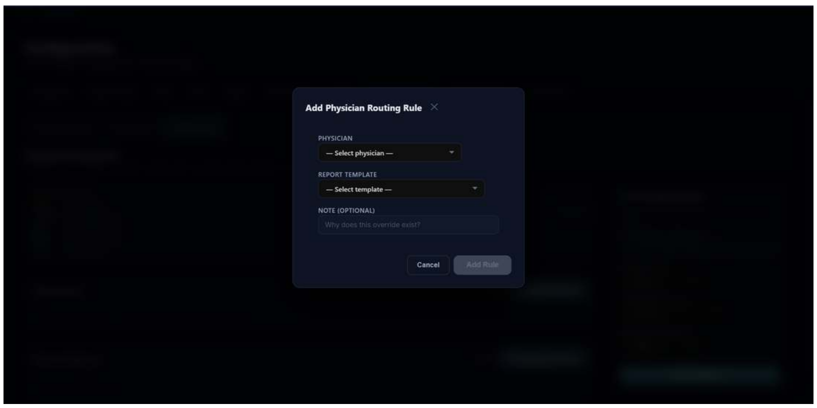
6.3 CAP Protocol Mappings

Sits between Client Overrides and Physician Preferences, matching its position in the priority chain (Pass 1). Lets you map a CAP/RCPath synoptic protocol directly to a Report Template without a code deploy -- for a newly-added protocol, or to override a mapping in the built-in table.

1. In CAP Protocol Mappings, click + Add Protocol Mapping.
2. Select the protocol (from the same live Synoptic Library picker used in Test Template Routing) and the Report Template it should resolve to.
3. Save. Takes effect immediately -- admin-defined mappings always take precedence over the built-in table for the same protocol ID.

6.4 Physician Preferences

Configured the same way as Client Overrides, matched against the case's ordering physician instead.



Add Physician Routing Rule -- select physician, target Report Template, optional note.

> **KNOWN LIMITATION (as of this writing)**
>
> Physician matching is currently keyed against the case's requestingProvider field, which elsewhere in the system is treated as a display name (e.g. "Dr. Jennifer Davis") rather than a stable physician ID. A Physician Preference rule defined here will only reliably match real cases once a stable physician-ID field exists on the case record. Until then, treat this pass as functionally correct but not yet dependably wired to live data.

6.5 Test Template Routing

A live simulation panel: enter a Synoptic Template ID, Subspecialty, Performing Client, and/or Ordering Physician, click Test Routing, and see exactly which pass matched (with a green checkmark) and why, with lower passes marked "Not reached" if a higher pass intercepted first.

> **NAMING NOTE**
>
> The on-screen field is currently labelled "CAP SYNOPTIC TEMPLATE ID," but it accepts any registered protocol ID -- CAP, RCPath, or Custom -- not CAP only. The same applies to Pass 1's on-screen name, "CAP protocol match." Both labels predate RCPath and Custom protocol support and are narrower than what the field/pass actually does; consider a more generic label (e.g. "Synoptic Template ID" / "Synoptic Protocol Match") in a future UI pass.

TIP

To deliberately exercise a specific pass during testing, leave the inputs for higher-priority passes empty. For example, to confirm Pass 1 (CAP Protocol Match) is wired correctly, leave Performing Client and Ordering Physician blank so Passes 0 and 0b have nothing to match against and fall through cleanly.

KNOWN LIMITATION (as of this writing)

There is no subspecialty field recorded directly on a case or specimen in the underlying data model. Pass 2 is fully implemented and can be exercised manually here by typing a subspecialty value directly, but it cannot yet be triggered automatically end-to-end from a real case. Treat Pass 2 as forward-looking until a subspecialty field is added upstream.

6.6 Known Limitations Summary

• Pass 2 (Subspecialty Fallback) is fully implemented and testable manually, but cannot yet be triggered automatically from a real case -- no subspecialty field exists on the case data model today.

• Pass 0b (Physician Preference) matches against a display-name field rather than a stable physician ID, and may not reliably match real cases until a proper physician-ID field exists.

7. How This All Connects to Real Report Generation

CORRECTED -- June 2026

This section previously described intended/target behavior that was not yet actually true: the pipeline below was disconnected from a separate, hardcoded narrative configuration (narrativeTemplateRegistry.ts) with no relationship to the real Parts/Assembly an admin configures in Sections 4-5. An admin could follow every step in this guide correctly and the real generated report would not reflect any of it. That gap was closed in June 2026 -- the description below is now accurate to the real, fixed pipeline, not aspirational.

Everything in Sections 4-6 is configuration. This section explains what actually happens when a real case is finalized and a report is generated.

7.1 The Generation Pipeline, Step by Step

1. A pathologist finalizes a case in Orchestration mode.
2. The Context Builder gathers the case's patient, accession, order, specimen, diagnostic, and coding data into a single validated structure.
3. The Context Builder asks Routing Rules to resolve which Report Template applies -- exactly the same five-pass logic described in Section 6, run automatically rather than via the Test panel.
4. The Context Builder fetches the resolved Report Template directly and walks its real, published Assembly -- every enabled Body Part, in order. Every Section within those Parts that has AI Generation enabled becomes a section the Orchestrator Engine will draft, using that section's configured AI instruction. This is the actual Template Assembly built in Sections 4-5 -- not a separate lookup.

5 The Orchestrator Engine drafts each AI-enabled section using the structured context and that section's AI instruction. Every other component in the Assembly (demographics, specimen lists, the synoptic summary table, sign-off) renders directly from case data -- no AI drafting involved -- respecting each component's Label Formatting and Visibility Condition exactly as configured in the Part Builder.

6 The pathologist reviews, edits as needed, and signs off.

[SCREENSHOT PENDING -- Pipeline diagram -- Case -> Context Builder -> Routing -> real Template Assembly -> Orchestrator -> Generated Report]

7.2 Why Configuration Accuracy Matters Here

Because routing resolution happens automatically as part of this pipeline, a misconfigured Routing Rule or an unpublished Report Template doesn't just affect the admin screens -- it changes what pathologists actually see when they generate a report, with no warning at generation time that something upstream was misconfigured. Three checks are worth making a habit of before considering any routing change "done":

- Confirm the target Report Template is Published, not Draft -- Routing Rules can still be saved pointing at a Draft template, but it will not be eligible for real use until published.
- Use the Test Template Routing panel (Section 6.4) to confirm the rule resolves the way you expect before trusting it for live cases.
- Re-test after any change to the underlying CAP protocol mapping table or subspecialty list, since those are shared, code-level mappings that affect every template, not just the one you were working on.

7.3 Narrative Sections Are Subspecialty-Specific

Each Report Template's AI-enabled sections come directly from that template's own assembled Parts -- the Breast Pathology Report's Diagnosis/Gross/Microscopic/Ancillary Parts carry breast-specific AI drafting instructions referencing ER/PR/HER2/Ki-67; the Urological Pathology Report's carry Gleason scoring instead; the Gastrointestinal Pathology Report's reference MSI/MMR testing conventions. This is why correct routing resolution matters beyond just picking the right header/footer -- an incorrectly-routed case will generate using the wrong subspecialty's narrative sections and AI instructions entirely, not just the wrong visual layout.

7.4 Sign-off and Audit

The signing pathologist's identity and credentials are captured as part of the same context used for generation, and key actions in the Synoptic Report workflow are recorded to an audit log independently of report content -- this is separate from, and unaffected by, anything configured in this guide.

8. Step-by-Step Workflows

8.1 Build a New Report Template for a New Subspecialty, End to End

1 Identify which existing Parts can be reused as-is (almost always: the standard headers, footers, Patient Demographics, Clinical Information, Specimens Submitted, Pathologist Sign-off) and which need to be new or customized for this subspecialty (almost always: Gross Description, Microscopic Description, Synoptic Summary, and any biomarker-specific section).

2 In Part Library, duplicate the closest existing body Part for anything that needs subspecialty-specific wording, and edit the duplicate (Section 4.10).

3 Publish each new/edited Part once it's ready (Section 4.9).

4 In Report Templates, click From Standard to start from the Gold Standard layout.

5. In the Body zone, remove any slots that don't apply to this subspecialty and add the subspecialty-specific Parts from Part Library, reordering as needed (Sections 5.4-5.5).
6. Click Preview and review the rendered output against mock data (Section 5.7).
7. Publish the template (Section 5.8).
8. In Routing Rules, confirm the CAP protocol mapping for this subspecialty already points at the new template's ID -- if this is a genuinely new subspecialty, this mapping needs to be added at the code level (Section 8.3) before Pass 1 will route to it automatically.
9. Use Test Template Routing to confirm resolution before considering the rollout complete (Section 6.4).

8.2 Add a Client Override So One Institution Always Gets a Specific Template

1. Confirm the target Report Template is already Published (Section 5.8).
2. In Routing Rules -> Client Overrides, click + Add Client Rule.
3. Select the client and the template.
4. Save.
5. In Test Template Routing, select that client (leave other fields as needed to confirm Pass 0 specifically) and click Test Routing. Confirm the result shows "Pass 0 -- Client override" with a green checkmark.

8.3 Add a New CAP Protocol Mapping (Code Change Required)

Unlike Client Overrides and Physician Preferences, the CAP protocol -> Report Template mapping used by Pass 1 is not editable from this screen -- it lives in code (TemplateRoutingService.ts). This workflow is for an administrator coordinating with engineering, not a self-service screen action.

1. Confirm the protocol's exact ID as registered in the Synoptic Library (visible in the Test Template Routing dropdown, Section 6.4) -- not a guessed or remembered version of the ID.
2. Confirm which Report Template it should map to.
3. Request the mapping be added, citing the exact protocol ID and target template ID.
4. Once deployed, verify with Test Template Routing using that exact protocol ID, with no Client/Physician override active, to confirm Pass 1 now resolves correctly.

> **WHY THIS MATTERS**
>
> Earlier in this system's lifecycle, this mapping table used IDs in a different naming format than the protocols actually registered in the Synoptic Library, which meant Pass 1 silently never matched any real case. This has since been corrected, but it illustrates why Step 1 above (confirming the exact real ID) matters -- a plausible-looking but incorrect ID will fail silently rather than producing an error.

8.4 Test a Routing Rule Before Trusting It for Real Cases

1. Open Routing Rules -> Test Template Routing.
2. Enter only the inputs relevant to the pass you want to verify, leaving higher-priority pass inputs blank so they fall through cleanly (Section 6.4 tip).

3. Click Test Routing.
4. Confirm the Resolution Priority list shows the expected pass highlighted with a green checkmark, and that its detail text matches what you intended (e.g. the correct client name and target template).
5. Repeat with different input combinations to confirm passes you don't want to fire (e.g. an unrelated client) correctly fall through instead of accidentally matching.

8.5 Create a New Reusable Part and Add It to Multiple Templates

1. In Part Library, click + New Part and choose the appropriate type.
2. Build out its content using the component palette (Section 4.5) and configure each component's Inspector settings (Section 4.6).
3. Publish the Part.
4. In each Report Template that should use it, open Template Assembly, click + Add on the relevant zone, and select the new Part from the Part Library sidebar (Section 5.4).

> **TIP**
>
> Because Parts are reused by reference, editing a published Part's content later updates every template that uses it the next time it's rendered -- keeping a body section like "Pathologist Sign-off" consistent across every template without having to update each one individually.

9. Settings & Fields -- Complete Reference

9.1 Part Builder -- General Inspector Fields (all component types)

Field	Type	Default
Label	Text	(component type name)
Width	Slider, 1-12	12 (full width)
Required	Toggle	Off
Hide if Empty	Toggle	Off
FHIR Export	Toggle	Off
Page Break Before	Toggle	Off
Label Formatting -- Position	Adjacent / Above / None	Adjacent (Above for paragraphs)
Label Formatting -- Transform	UPPERCASE / Capitalize / None	UPPERCASE
Label Formatting -- Weight / Decoration / Size	Bold toggle / Underline toggle / px	Normal / None / 11-12px
Visibility Condition	Expression Builder (AND/OR clauses)	No clauses -- always shown

9.2 Part Settings Panel

Field	Type
Type	Header / Body / Footer
Specialty	Free text
Description	Free text

Field	Type
Standard	CAP / RCPath / Custom

9.3 Template Assembly Screen

Control	Effect
Template name (topbar)	Renames the template; click pencil/hover to edit
+ Add [zone]	Activates that zone for Part picking from the sidebar
Drag handle icon	Reorders Body slots only
Enable/disable dot	Skips a slot during generation without removing it
Edit part	Opens the Part Builder for that slot's Part
Remove	Removes the slot; the Part itself is unaffected
Preview	Renders the template against mock data
Publish	Draft -> Published; one-directional

9.4 Routing Rules -- Test Template Routing Panel

The on-screen field name says CAP specifically; see the Naming Note in Section 6.5 -- it accepts RCPath and Custom protocols equally.

Field (as labelled on screen)	Source	Feeds
CAP Synoptic Template ID	Live picker -- real Synoptic Library registry (CAP, RCPath, or Custom)	Pass 1
Subspecialty	Free text	Pass 2
Performing Client	Live picker -- real client list	Pass 0
Ordering Physician	Live picker -- real physician list	Pass 0b

9.5 Status Badges (Parts and Report Templates)

Badge	Meaning
Draft	Not yet eligible for routing or real generation
Published	Live -- eligible for routing and real generation
Archived	Retired; retained for historical/audit integrity, not offered for new use
BUILT-IN	Protected Part; not editable or archivable, only duplicable
AI	Orchestrator mode is enabled for this template
STANDARD	This is the designated Gold Standard template (Pass 3 fallback)

10. Troubleshooting & FAQ

"A section of my Part doesn't appear in Preview."

Check, in order: (1) whether "Hide if Empty" is enabled on that component and its bound field has no value in the mock context -- try the Context JSON panel in Preview (Section 5.7) to confirm what data is actually available; (2) whether a Visibility Condition is configured on the component that isn't satisfied by the mock context; (3) whether the binding key is spelled exactly as it appears elsewhere -- binding keys are case-sensitive and a typo resolves silently to nothing rather than producing an error.

"My Routing Rule isn't matching when I test it."

Check, in order: (1) whether a higher-priority pass is intercepting first -- use the Resolution Priority list's per-pass detail text (Section 6.4) to see exactly what each pass actually evaluated, rather than assuming; (2) whether the rule is marked Active; (3) whether the target Report Template is Published, not Draft; (4) for a Client Override or Physician Preference, whether you're testing with the exact same client/physician the rule was defined against.

"I renamed a Part, but the Report Template still shows the old name."

This is a known limitation (Section 4.11 / 5.9). Remove the slot from the template and re-add it from Part Library to refresh the displayed name. The actual rendered content was never affected -- only the name shown in the Assembly screen goes stale.

"I don't see my latest edit in Preview."

Check the auto-save indicator in the topbar (Section 4.9). If it still shows "Unsaved" or "Saving...," wait a moment for it to settle to "Saved" before opening Preview.

"Why does the Resolution Priority list show some passes as 'Not reached'?"

Routing is a priority cascade, not an independent checklist (Section 6.1) -- once a pass matches, lower-priority passes are genuinely never evaluated, not just hidden from the result. "Not reached" accurately reflects that the resolver stopped before getting to that pass, not that it was checked and happened to fail.

"Can I revert a Published Part or Template back to Draft?"

Not directly -- publishing is one-directional in the current interface (Sections 4.9, 5.8). The standard workaround is to duplicate the published item, make changes to the duplicate, and re-point usage at the duplicate once it's ready.

"Why is the CAP Synoptic Template ID field a dropdown instead of a text box?"

It represents a Synoptic Template ID (Section 2.4) -- despite the on-screen name, any CAP, RCPath, or Custom protocol is valid here, not CAP only (see the Naming Note in Section 6.5). It's populated live from the real Synoptic Library registry specifically so that every option offered is guaranteed to be a real, currently-recognized ID. A free-text field here would allow entering a plausible-looking but incorrect ID, which would silently fail to match rather than producing a visible error -- exactly the class of mistake this design choice exists to prevent.

"Which screen do I use to change what data feeds a CAP/RCPath synoptic checklist?"

None of the screens in this guide -- that's the Synoptic Library section of Configuration, a separate system from Report Templates (Section 2.4). This guide covers only the document shell and its routing, not the underlying synoptic data-collection checklists themselves.

Appendix A: Built-in (Protected) Parts

All built-in Parts shipped with PathScribe. None are editable directly -- duplicate first to customize (Section 4.10).

Name	Type	Description
PathScribe Standard Header -- Page 1	Header	Full institutional header for page 1: logo, branding, accession, and patient demographics.
PathScribe Compact Header -- Pages 2+	Header	Compact running header for pages 2+: accession, patient, and DOB on one line.
PathScribe Standard Footer -- Page 1	Footer	Standard page-1 footer including patient identity information.
PathScribe Compact Footer -- Pages 2+	Footer	Compact footer for pages 2+.
Patient & Order Demographics	Body	Auto-populated patient and order demographics.
Clinical Information	Body	Clinical indication auto-populated from the request, with an optional free-text addition.
Specimens Submitted	Body	Auto-populated list of all submitted specimens with site, type, and fixative.
Diagnosis	Body	AI-drafted diagnosis per specimen; pathologist reviews and confirms.
Synoptic Summary	Body	Auto-populated synoptic data table -- two-column primary findings and biomarkers. No pathologist input.
Gross Description	Body	AI-generated gross description per specimen; pathologist reviews.
Microscopic Description	Body	AI-generated microscopic description per specimen; pathologist reviews.
Ancillary Studies	Body	AI-summarised IHC and molecular results. Conditional -- only appears if IHC was performed.
Comment	Body	Optional free-text comment block.
Pathologist Sign-off	Body	Pathologist sign-off / attestation block.

Appendix B: Seeded CAP / RCPath Synoptic Protocols

The synoptic protocols currently registered in the Synoptic Library, organized by subspecialty. These are the IDs the CAP Protocol Match pass (Section 6.1, Pass 1) recognizes, and the same list the Test Template Routing picker (Section 6.4) is populated from.

Subspecialty	Protocol ID	Source
Breast	breast_invasive	CAP
Breast	breast_dcis_resection	CAP
Breast	rcpath_g148_breast_surgical_excision	RCPath
Gastrointestinal	colon_resection	CAP
Gastrointestinal	rcpath_colorectal_resection	RCPath
Gastrointestinal	rcpath_colorectal_local_excision	RCPath

Subspecialty	Protocol ID	Source
Gastrointestinal	rcpath_colorectal_further_investigations	RCPath
Thoracic / Pulmonary	lung_adeno	CAP
Thoracic / Pulmonary	lung_resection	CAP
Urological	prostate_needle_biopsy	CAP
Urological	prostate_resection	CAP
Urological	rcpath_prostate_biopsy	RCPath
Urological	rcpath_prostate_radical_prostatectomy	RCPath
Urological	rcpath_prostate_turp_enucleation	RCPath
Urological (renal)	kidney_resection	CAP
Urological (renal)	kidney_biopsy	CAP
Urological (renal, paediatric)	wilms_resection	CAP
Urological (renal, paediatric)	wilms_biopsy	CAP
Dermatopathology	skin_melanoma_bx	CAP
Dermatopathology	skin_invasive_melanoma_biopsy	CAP

NOTE

Dermatopathology protocols currently route to the Gold Standard Report Template -- there is no dedicated Dermatopathology Report Template as of this writing. The same is true of any protocol not listed above; it will route via Pass 2 (Subspecialty Fallback) or Pass 3 (Gold Standard) rather than Pass 1.

Appendix C: Glossary

Term	Definition
Assembly Slot	One Part placed into a zone (Header/Body/Footer) of a Report Template; records the Part used, its role, order, and enabled state.
Binding Key	The dot-path (e.g. order.fullAccession) a component uses to pull its value from case data.
Cascade	The routing behavior where passes are tried in strict order and evaluation stops at the first match.
Component	Any item from the Part Builder palette (Text Field, Section, If/Show When, etc.) placed on the canvas.
Gold Standard	The one Report Template designated as the universal fallback (Pass 3); always matches if nothing else does.
Inspector	The right-hand panel in the Part Builder showing settings for the currently selected component.
Orchestrator	The engine that resolves the correct Report Template and AI-drafts narrative sections for a finalized case.
Part	A reusable Header, Body, or Footer building block (Section 2.1).
Report Template	A complete document shell assembled from Parts (Section 2.2).

Term	Definition
Resolution Pass	One step in the five-step routing priority chain (Section 2.5, 6.1).
Routing Rule	An administrator-defined override forcing a specific template for a client or physician (Section 2.3).
Synoptic Template (CAP / RCPath / Custom)	A structured data-entry checklist for one specimen type, configured in the Synoptic Library -- distinct from a Report Template (Section 2.4).
Visibility Condition	An Expression Builder condition controlling whether a component renders at all.

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